

Level, Sharpness, and Corpus: Why Zero-Shot OOD Detector Rankings Do Not Transfer

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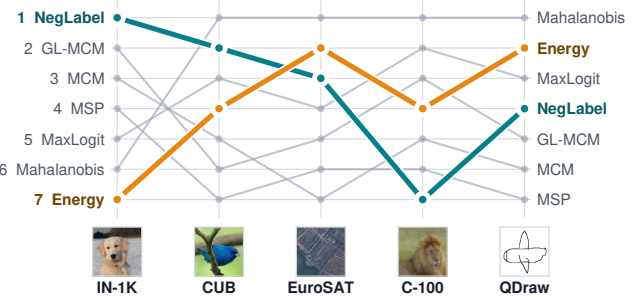
Abstract

Selecting a zero-shot out-of-distribution (OOD) detector for a new deployment is typically based on benchmark rankings, implicitly assuming that the highest-ranked detector will transfer across domains. We show that this assumption does not hold. Through a controlled portability audit across seventeen in-distribution datasets, three vision-language models, and seven representative zero-shot OOD detectors, we find that detector rankings reverse across deployments, every detector exceeds 80% FPR95 on at least one domain, and the preferred detector depends on both the in-distribution data and the underlying VLM. We trace these reversals to complementary evidence channels in vision-language logits. Corpus-free detectors rely on different combinations of absolute match level and relative or spatial sharpness, while WordNet-based methods additionally depend on external semantic coverage. A simple proposition shows that level and sharpness cannot generally be recovered from one another, explaining why no single detector transfers reliably across deployments. Motivated by this diagnosis, we introduce the Complementary Evidence Guard (CEG), a detector-agnostic wrapper that preserves complementary evidence through a non-compensatory fusion of the base detector, level, and sharpness using only empirical in-distribution percentiles. Controls replacing these channels with entropy, logit variance, or random noise do not reproduce the gains. Without OOD samples, auxiliary corpora, or learned fusion, CEG reduces detector sensitivity and improves GL-MCM from 38.1 to 28.8 and MCM from 42.6 to 30.5 family-balanced FPR95.

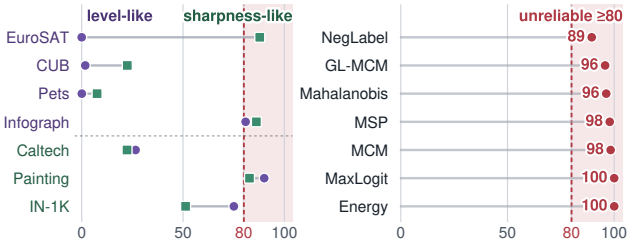
1 Introduction

Frozen vision-language models (VLMs) have made zero-shot out-of-distribution (OOD) detection practical across a wide range of applications. Given an image and a set of in-distribution (ID) class names, a detector can estimate whether the image belongs to the ID distribution without task-specific retraining (Radford et al. 2021; Ming et al. 2022). The remaining challenge is deciding which detector should be deployed. Existing methods are typically ranked on a benchmark, and the highest-performing score is then reused in new deployment settings. Figure 1 shows that this practice is unreliable. Across a seventeen-dataset CLIP audit, detector rankings reverse repeatedly as the ID domain changes. The same conclusion holds across three vision-language models in a separate cross-VLM portability study. The detectors

(a) Score rankings reverse



(b) Best evidence flips



(c) Every score fails

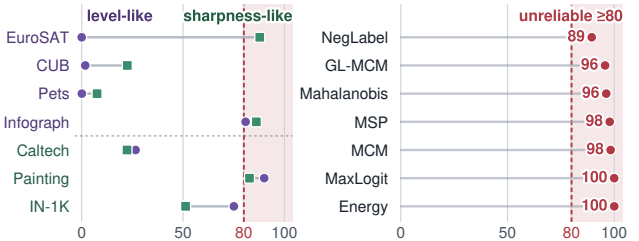


Figure 1: Benchmark winners do not transfer across ID domains. (a) Detector rankings reverse across ID domains. (b) Different ID domains favour different evidence. (c) Every detector fails on at least one ID domain, showing that no single zero-shot score is universally reliable.

themselves remain valid. What fails is the assumption that benchmark rankings transfer across deployments.

This assumption has become widespread because benchmark evaluation has driven much of the recent progress in zero-shot OOD detection. Methods are commonly compared on ImageNet-centred benchmarks, where improvements are often reported as a single average across ID–OOD shifts. Consequently, the detector with the best benchmark performance is frequently treated as the preferred deployment choice. Previous work questioned whether these benchmarks faithfully reflect deployment because of dataset contamination and benchmark bias (Bitterwolf, Müller, and Hein 2023; Fort, Ren, and Lakshminarayanan 2021). We ask a different question. Even if the benchmark itself is valid, does the detector ranking obtained on one ID domain remain valid when the ID domain changes?

To answer this question, we conduct a controlled portability study while holding implementations, preprocessing, and evaluation protocols fixed. Across seventeen ID domains, three VLMs, and seven representative post-hoc OOD detectors, rankings reverse repeatedly as either the ID domain or the underlying VLM changes. Every detector exceeds 80% FPR95 on at least one ID domain, and no method remains consistently superior. The problem is therefore not simply identifying the detector with the best average benchmark performance, but understanding why different detectors succeed on different ID domains.

Figure 1 suggests the underlying mechanism. Existing detectors rely on complementary evidence channels. *Level* measures how strongly an image matches the ID classes in absolute terms, whereas *sharpness* measures how clearly one class stands out relative to the remaining vocabulary and spatial locations. Some methods additionally exploit external semantic coverage. Different ID domains preserve different combinations of these signals, making no single evidence channel universally sufficient. This perspective explains why detector rankings reverse across ID domains and suggests preserving complementary evidence rather than relying on a single score. Instead of seeking a universally best detector, the goal becomes reducing deployment sensitivity by retaining evidence that individual scores discard, leading to more stable decisions across ID domains.

This perspective changes the deployment objective. If different ID domains require different evidence, selecting a single benchmark winner is no longer sufficient. Instead, deployment should preserve complementary evidence so that decisions remain reliable when the informative evidence changes across ID domains. Guided by this principle, we propose the *Complementary Evidence Guard* (CEG), a detector-agnostic framework that calibrates complementary evidence using unlabeled ID data and combines it through a non-compensatory decision rule. Rather than replacing existing detectors, CEG is designed to make them more robust to changes in the ID domain by preserving the evidence that individual scores discard.

Contributions.

- **A portability audit.** We show that zero-shot OOD detector rankings do not transfer across ID domains or VLM families.
- **An evidence-level explanation.** We explain ranking reversals through complementary evidence channels whose utility depends on the ID domain.
- **A complementary evidence guard.** We introduce a detector-agnostic wrapper that preserves complementary evidence without retraining.
- **A deployment-oriented evaluation protocol.** We advocate reporting worst-domain risk and revalidating detectors whenever the ID domain or VLM changes.

In short. Benchmark winners are not necessarily deployment winners. Different ID domains preserve different evidence, making complementary evidence more reliable than relying on a single detector.

2 Related Work

Post-hoc OOD detection. Classical detectors threshold classifier confidence: maximum softmax probability (Hendrycks and Gimpel 2017), maximum logit (Hendrycks et al. 2022), free energy (Liu et al. 2020), or Mahalanobis distance (Lee et al. 2018). VLM variants apply related statistics to image-text logits: MCM uses the ID-vocabulary softmax (Ming et al. 2022); GL-MCM adds a local term (Miyai et al. 2025); and NegLabel adds WordNet negatives (Jiang et al. 2024; Miller 1995). ZOC, CLIPN, NegPrompt, and LoCoOp instead add generated negatives, a learned “no” embedding, negative prompts, or ID prompt adaptation (Esmailpour et al. 2022; Wang et al. 2023; Li et al. 2024; Miyai et al. 2023). LogitGap contrasts selected logits (Liang et al. 2025). Conjugated Semantic Pool (CSP) mines external WordNet noun/adjective prompts and scores positive mass against that expanded pool (Chen, Gao, and Xu 2024). Later variants robustly weight negatives (Peng et al. 2026) or adapt proxies at test time (Zhang and Zhang 2024). Large-scale classifier studies also expose the limits of single-benchmark conclusions (Fort, Ren, and Lakshminarayanan 2021). Most VLM evidence remains ImageNet-centered; we vary the ID dataset and isolate the external corpus’s role.

Resource assumptions. MSP, MaxLogit, Energy, and MCM need only the ID vocabulary; GL-MCM also reads patches (Ming et al. 2022; Miyai et al. 2025); Mahalanobis fits ID statistics (Lee et al. 2018); and NegLabel, CSP, and AdaNeg add external semantics (Jiang et al. 2024; Chen, Gao, and Xu 2024; Zhang and Zhang 2024). Prompt learning adds adaptation. We keep these resource tiers separate after wrapping; CEG calibrates its channels to ID percentiles before taking their minimum.

Benchmarks and VLM portability. OpenOOD standardizes ImageNet-scale evaluation (Yang et al. 2022; Zhang et al. 2023). Prior work varies CLIP encoders, and NegLabel/EOE also test ALIGN, GroupViT, or AltCLIP, but keep ImageNet-1K and its usual OOD suite (Ming et al. 2022; Jiang et al. 2024; Cao et al. 2024). We instead test score-rank invariance while changing both VLM and ID domain.

3 The Portability Audit

Zero-shot OOD work primarily develops and evaluates methods on CLIP and ImageNet, the dominant benchmark in the vision-language OOD literature, before deploying the same formula on different wID domains or VLMs. This implicitly assumes that a runnable detector retains its empirical rank. We begin by testing that assumption.

A benchmark winner is not a deployment winner. If benchmark rankings were portable, the detector with the best benchmark average should also be the safest deployment choice. Figure 2c shows that this is not the case. Every fixed score exhibits a large gap between its mean and worst-domain performance. Across the complete seventeen-domain CLIP study (Appendix Table B1), every detector exceeds 80 FPR95 on at least one deployment, and the best worst case is still 89. Selecting the lowest benchmark mean therefore does

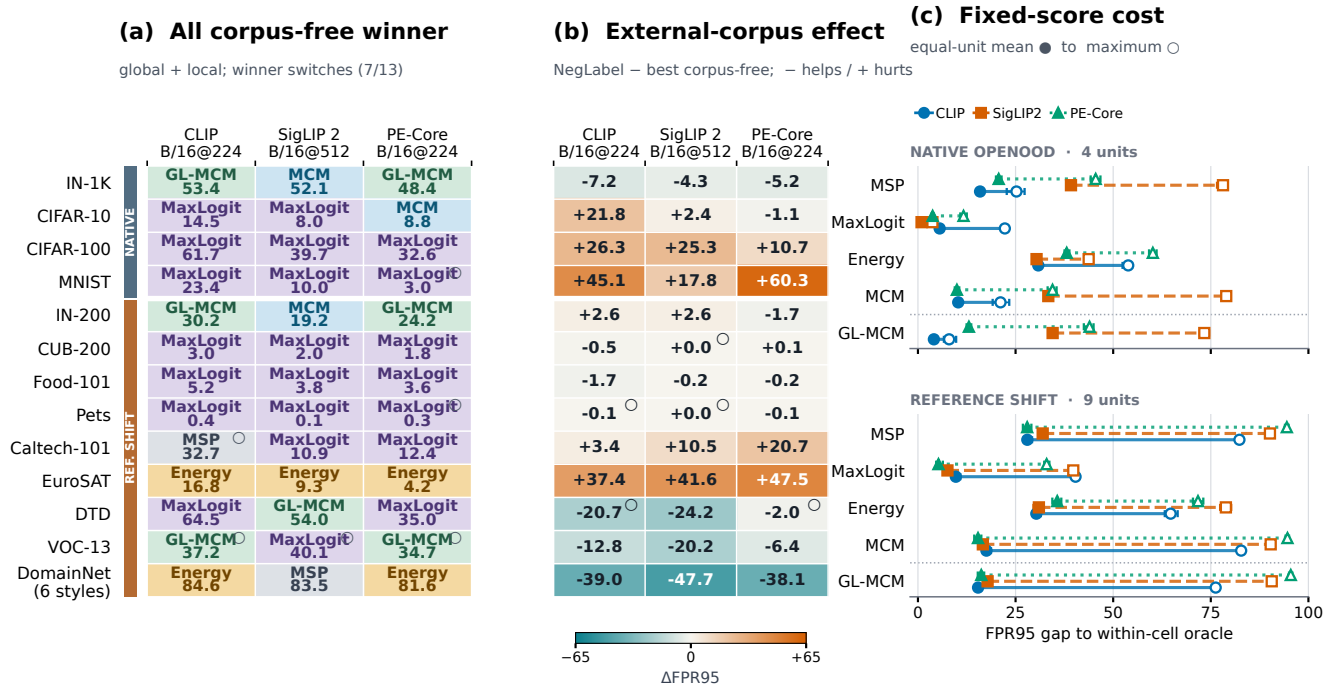


Figure 2: **Portability fails because detectors preserve different evidence.** (a) The best corpus-free detector changes across ID domains. (b) The contribution of external semantic coverage depends on the ID domain and VLM. (c) The cost of fixing one score: filled markers show the average FPR95 gap to the best score for each unit, and open markers show the worst-case gap. Farther right means worse transfer.

not control the failures practitioners encounter after the ID domain changes.

To distinguish failures caused by the ID domain from those introduced by the benchmark itself, we evaluate two complementary deployment settings. *Native* evaluations use each dataset’s official OpenOOD-v1.5 benchmark and OOD suite (Zhang et al. 2023), whereas *reference-shift* evaluations score different ID datasets against a shared ImageNet-based OOD reference. Together these separate changes caused by the deployment domain from those introduced by the benchmark. Complete implementation details are given in App. B.

The model changes the answer too. Figure 2a shows that replacing only the underlying VLM also changes the preferred detector. The all-five winner changes in 7/13 deployment units (two native and five reference-shift), while the frozen global subset still changes in 5/13 units after controls. Winner probability exceeds 0.95 in 33/39 all-method cells, so these reversals are not merely close statistical ties. Neither the deployment domain nor the representation alone determines a universally best detector, suggesting that the answer instead depends on the evidence each detector preserves.

The failures are complementary. Figure 2b explains why rankings reverse. NegLabel succeeds when WordNet brackets the deployment shift but fails on CIFAR-like domains, whereas Mahalanobis benefits from compact ID feature clusters but deteriorates on broad ImageNet deployments. Their nearly disjoint strengths indicate that domains do not reward

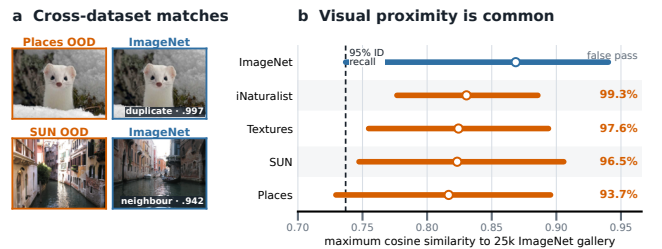


Figure 3: **Benchmark composition is part of the audit.** (a) A verified Places/ImageNet duplicate and a selected SUN neighbour illustrate that label-defined OOD need not be visually disjoint. (b) Maximum-cosine distributions overlap substantially between ImageNet and the traditional OOD sources. Full protocol: Appendix C.

universally better detectors; they reward different evidence channels. This naturally raises the central question: what evidence do existing detectors actually preserve? We answer that question in Section 4.

Benchmark composition is part of the audit. Portability should be evaluated independently of artefacts introduced by the benchmark itself. Traditional ImageNet protocols define far-OOD using iNaturalist, SUN, Places, and Textures (Huang and Li 2021), although label-defined OOD need not be visually or data disjoint. Figure 3 shows both a verified Places/ImageNet duplicate and substantial over-

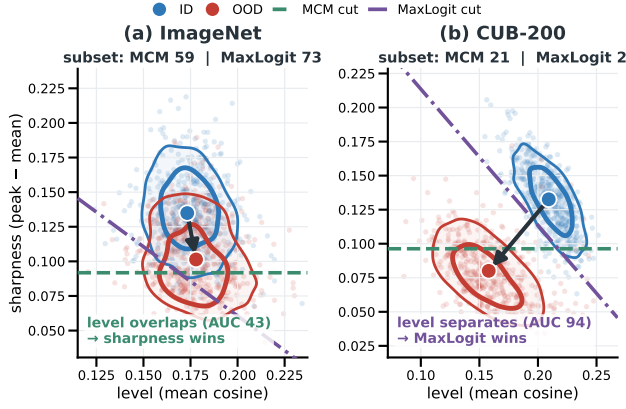


Figure 4: **Level and sharpness separate different shifts.** Sharpness separates ImageNet, whereas absolute match level separates CUB-200, explaining the MCM/MaxLogit reversal. Contours and header FPR95 use a diagnostic subset (20 ID images/class; at most 3,000/OOD source); Tables 1 and B1 report separate protocols. Horizontal position is mean cosine, the level term in MaxLogit.

lap between the corresponding maximum-cosine distributions, where 93.7–99.3% of nominally OOD images pass the ImageNet 95% recall threshold. This probe does not estimate contamination prevalence, but it shows that the traditional benchmark is not visually clean. We therefore adopt OpenOOD-v1.5, which explicitly separates near- and far-OOD shifts while avoiding these known benchmark issues (Zhang et al. 2023; Bitterwolf, Müller, and Hein 2023).

4 Mechanism: Level and Sharpness Are Complementary Evidence

The portability audit showed that detector rankings reverse because different ID domains reward different evidence channels. We now ask what that evidence is. Figure 4 shows that CLIP cosine logits contain two partially independent signals:

- the **level** $\max_c \ell_c$: how strongly the image matches its best anchor in absolute terms;
- the **sharpness** $\max_c \ell_c - \text{mean}_c \ell_c$: how much the best match stands out from the remaining vocabulary and, spatially, how much the strongest image region stands out from the rest of the image.

Different samples can fail through different channels. An image may strongly match the ID vocabulary in absolute terms yet lack a coherent class peak, whereas another may produce a sharp winning class despite remaining atypical in absolute level. A single score therefore cannot capture every failure, motivating the non-compensatory guard introduced next.

How existing detectors use these channels. The previous analysis explains why both level and sharpness are needed. Existing detectors, however, do not preserve both. Instead, each detector implicitly commits to one form of evidence, which explains why their rankings reverse across ID domains.

MCM discards level. The softmax formulation of MCM obscures a simple but important fact.

Proposition 1. $S_{\text{MCM}}(x) = \max_c \text{softmax}(\ell/T)_c$ is invariant to additive logit shifts $\ell_c \mapsto \ell_c + \gamma(x)$, and for $T \gg R$ (logit range), $S_{\text{MCM}} = \frac{1}{K} + \frac{1}{KT} (\max_c \ell_c - \text{mean}_c \ell_c) + O(R^2/T^2)$.

Proposition 1 identifies MCM as a shift-invariant approximation to the centred maximum, thereby preserving only the sharpness channel. The complementary decomposition follows from the log-sum-exp identity. Writing $\ell_c = \bar{\ell} + \delta_c$ with $\sum_c \delta_c = 0$,

$$\text{Energy}(\ell) = \log \sum_c e^{\ell_c} = \log K + \bar{\ell} + \frac{\text{Var}(\ell)}{2} + O(\|\delta\|^3),$$

thus, Energy preserves the absolute level to leading order, with only a second-order dependence on logit variance. Empirically, both approximations are essentially exact in our setting (Appendix E.3): MCM differs from the centred maximum by less than 0.2 FPR95 across all core benchmarks, while Energy closely matches the mean level. Together, MCM and Energy form two complementary extremes of the same logit decomposition: one preserves sharpness, whereas the other preserves level.

This decomposition makes the portability failures interpretable. ID domains separated primarily by sharpness favour MCM-like scores, whereas deployments separated primarily by level favour Energy- or MaxLogit-like scores. Because the informative component changes with the ID domain, neither family can retain a universal ranking.

This immediately explains the behaviour observed in Figure 4. On ImageNet, a 1,000-anchor vocabulary gives ID images one sharp winner while far-OOD matches nothing sharply. On EuroSAT and Quickdraw, however, ID images themselves match no anchor sharply (satellite tiles and line drawings are far from CLIP’s photographic text concepts), so the sharpness channel contains little ID/OOD information and MCM approaches chance performance. Conversely, on CUB-200 the *level* almost perfectly separates ID from OOD, yet shift-invariant scores discard that signal, giving MCM 23 versus MaxLogit 2.

Vocabulary size confounds the score. Increasing the number of anchors K changes both maxima and normalisers, so part of ImageNet’s apparent ranking is the vocabulary rather than the images. In controlled class subsets (Figure 5a), MaxLogit pays a growing penalty to MCM. Growing the label set alone can therefore change detector behaviour.

Each detector commits to different evidence. The previous analysis explains the behaviour of the corpus-free detectors. MSP and MCM preserve sharpness because softmax normalisation discards the level; MaxLogit, Energy, and Mahalanobis preserve the level (or its feature-space analogue); and GL-MCM extends sharpness with local evidence. The portability audit therefore becomes interpretable: level-separable domains (CUB, Pets, Food, EuroSAT, Quickdraw, and MNIST) reward MaxLogit and Mahalanobis, whereas

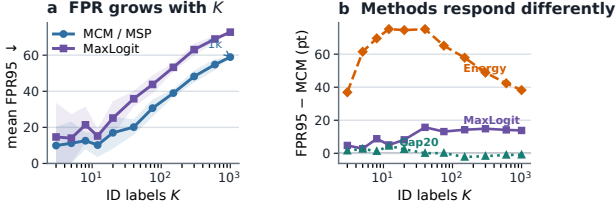


Figure 5: **Label-set construction changes the detector.** (a) Mean FPR95 on a fixed suite under ImageNet class subsampling (eight draws; bands ± 1 s.d.). (b) Other global logit scores minus MCM; positive favours MCM, and MSP=MCM under this scoring convention.

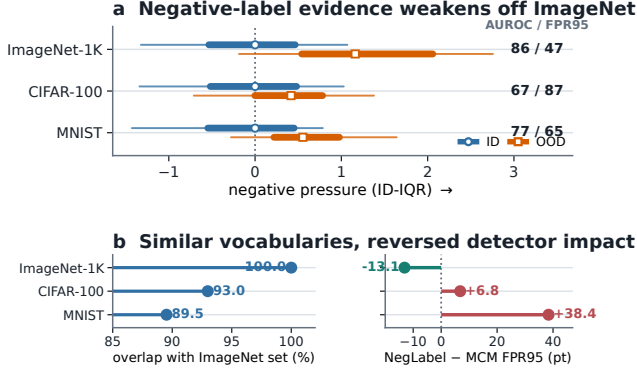


Figure 6: **External semantic evidence depends on corpus coverage.** (a) Negative-label evidence separates ImageNet but weakens off ImageNet. (b) NegMining selects nearly the same WordNet vocabulary across deployments, yet detector performance reverses. Corpus coverage, not vocabulary selection, limits transfer.

sharpness-separable domains (ImageNet-1K/200 and Caltech) reward MCM and GL-MCM. Domains where neither channel is reliable (Painting and Infograph) leave every corpus-free detector at or above 49 FPR95.

This leaves one remaining family of detectors. NegLabel and CSP do not rely solely on the ID vocabulary. Instead, they augment it with a WordNet-derived semantic corpus, introducing a third source of evidence that depends on the coverage of that external vocabulary rather than only on the image representations themselves.

4.1 External Semantic Evidence Depends on Corpus Coverage

Unlike the corpus-free detectors, NegLabel and CSP augment the ID vocabulary with WordNet-derived concepts. Their success therefore depends not only on the image evidence, but also on whether this external semantic corpus brackets the ID–OOD shift. Figure 6 asks two related questions: *does the mined vocabulary provide useful OOD evidence, and does NegMining adapt that vocabulary when the ID domain changes?*

Panel (a) shows that the external semantic evidence separates ID and OOD on ImageNet but rapidly loses discrimina-

tive power on CIFAR-100 and MNIST. Panel (b) explains this reversal. Although the ID domain changes substantially, NegMining selects almost the same WordNet vocabulary, with 89.5 to 93.0% overlap with the ImageNet negatives. Nevertheless, NegLabel changes from improving MCM by 13.1 FPR95 points on ImageNet to degrading it by 6.8 and 38.4 points on CIFAR-100 and MNIST, respectively. The limitation therefore lies not in the mining procedure, but in the coverage of the external semantic corpus. WordNet brackets ImageNet-style shifts well, yet provides much weaker evidence for other deployment domains. CSP relies on the same WordNet corpus and therefore inherits the same dependence on corpus coverage.

The audit identifies three complementary evidence channels. Because deployment domains preserve different channels, no fixed detector transfers reliably. This diagnosis naturally motivates CEG, which preserves complementary evidence rather than relying on a single score.

5 From Diagnosis to Deployment: The Complementary Evidence Guard

The portability audit and its evidence-level explanation suggest a different deployment strategy. Rather than selecting the detector that performed best on a benchmark, deployment should preserve whichever evidence remains informative after the domain changes. This leads naturally to the Complementary Evidence Guard (CEG), which preserves complementary evidence using only unlabeled ID calibration data.

Complementary channels. For a higher-is-ID base detector B and unlabeled ID calibration set $\mathcal{C} = \{x_j\}_{j=1}^n$, the same VLM forward pass produces two complementary channels,

$$L(x) = z_{\text{ID}}\left(\max_c \ell_c\right) + z_{\text{ID}}\left(\frac{1}{k} \sum_{i \in \mathcal{T}_k(x)} \max_c \ell_{i,c}\right),$$

$$S(x) = z_{\text{ID}}\left(\max_c \ell_c - \text{mean}_c \ell_c\right) + z_{\text{ID}}\left(\max_q (\kappa * s)_q\right). \quad (1)$$

Here $\mathcal{T}_k(x)$ contains the $k = 10$ patches with the highest relative confidence, so L combines global and local absolute level, whereas S combines global and spatial sharpness. Both channels are standardised using ID statistics estimated from $\mathcal{C}$.

Complementary Evidence Guard. Each channel is mapped to its empirical ID percentile,

$$U_R(x) = \hat{F}_R(R(x)),$$

where $\hat{F}_R$ is the empirical cumulative distribution function of channel R estimated from $\mathcal{C}$. The guard is then

$$G_B(x) = \text{CEG}(B)(x) = \min_{R \in \{B, L, S\}} U_R(x). \quad (2)$$

A sample is accepted whenever $G_B(x) \geq \tau$, where τ is fixed using a disjoint ID operating split (Appendix B).

Table 1: **CEG: strict results and full-domain coverage.** (a) Raw→CEG FPR95↓/AUROC↑ on strict tasks. (b) Every entry is $\Delta\text{FPR}/\Delta\text{AUC}$ (CEG−Raw): negative ΔFPR and positive ΔAUC denote improvement. Colors are metric-specific. In panel (a), each subcolumn is keyed independently and only regressions are tinted; in panel (b), the two numbers are colored independently. Green/red/gray denotes improvement/regression/ $|\Delta| \leq 0.10$ tie. DomainNet counts once in the 13-unit mean. Aggregation sensitivity is in Appendix Table E6.

(a) Strict five-task headline: Raw → CEG														
Base detector	IN-1K		IN-200		C-10		C-100		MNIST		Family			
	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑		
Class-name scores; no external corpus														
MaxLogit	72.7→54.3	77.3→82.3	45.7→30.2	89.3→92.2	14.5→11.7	96.2→97.2	61.5→52.6	85.7→87.5	23.5→15.8	95.1→97.1	40.2→30.1	89.8→92.2		
Energy (τ=1)	97.0→56.9	42.7→79.3	95.6→33.0	46.6→90.8	68.8→13.4	70.5→96.8	77.9→57.2	64.4→86.6	34.4→16.0	90.3→97.0	68.0→32.1	67.5→91.3		
MCM	60.0→53.9	81.9→82.9	32.2→28.9	91.7→92.5	19.4→11.5	94.8→97.2	82.4→56.8	76.6→85.9	30.8→16.0	93.5→97.0	42.6→30.5	88.7→92.1		
GL-MCM†	53.1→52.7	83.2→82.9	28.7→27.3	92.0→92.5	16.4→11.1	95.4→97.1	67.8→53.6	80.0→86.4	31.2→14.0	93.0→97.4	38.1→28.8	89.4→92.3		
20%-nonmax gap (abl.)	59.1→53.2	82.7→83.2	29.2→27.3	92.4→92.8	34.6→12.3	92.8→97.0	88.0→57.9	76.3→85.9	76.7→19.9	78.8→95.8	60.7→31.8	83.7→91.7		
LogitGap	59.1→53.2	82.7→83.2	29.2→27.2	92.4→92.8	26.5→11.8	93.9→97.1	88.0→57.9	76.3→85.9	50.7→18.9	87.5→96.2	50.7→31.3	86.7→91.9		
Unlabeled-ID feature fit														
Mahalanobis	96.6→57.3	62.1→80.5	86.9→32.9	82.0→91.8	18.1→8.2	95.9→98.0	37.7→31.3	90.9→93.4	0.2→0.4	99.9→99.8	40.0→21.7	88.5→93.9		
External WordNet corpus/mining														
NegLabel	46.2→46.0	85.9→85.0	29.3→26.1	93.0→93.2	36.1→11.9	91.9→97.1	88.7→59.4	66.7→85.0	69.0→15.8	76.4→96.9	56.4→29.2	81.7→92.3		
CSP	41.5→42.3	87.4→86.2	25.2→24.7	93.9→93.7	30.2→11.7	94.2→97.1	81.1→54.9	74.7→86.0	56.7→15.4	84.6→97.0	48.6→27.4	86.6→92.8		
(b) Every pitfall-audit unit: ΔFPR/ΔAUC (CLIP-B/16)														
Base	IN-1K	C-10	C-100	MNIST	IN-200	CUB	Food	Pets	Cal	Euro	DTD	VOC	DN	13-unit
MSP	-12.4/+4.2	-14.3/+3.3	-29.2/+10.6	-21.1/+6.5	-7.7/+1.9	-74.7/+30.1	-15.8/+3.2	-37.2/+8.4	-5.4/+2.1	-34.1/+36.3	-25.1/+16.1	-6.4/+4.7	-5.0/+5.3	-22.2/+10.2
MaxLogit	-21.1/+5.4	-3.4/+0.9	-8.9/+1.8	-8.0/+1.9	-14.4/+2.5	+0.3/-0.1	-0.5/-0.1	0.0/-0.1	-16.6/+4.4	+0.6/+1.4	-6.4/+2.7	-15.3/+5.0	-3.6/+9.3	-7.5/+2.7
Energy	-39.1/+38.2	-55.5/+26.3	-21.1/+22.4	-18.5/+6.7	-60.6/+40.5	-15.7/+4.5	-47.1/+11.9	-17.7/+4.8	-61.5/+42.1	+7.3/-2.5	-16.7/+8.8	-47.4/+33.4	-5.0/+25.1	-30.7/+20.2
MCM	-6.9/+1.0	-8.3/+2.4	-25.4/+9.5	-16.2/+3.6	-5.3/+1.2	-22.1/+4.3	-3.7/+0.7	-7.0/+1.4	-9.5/+2.1	-34.6/+47.1	-25.7/+13.9	-3.3/+3.1	-5.1/+4.8	-13.3/+7.3
GL-MCM†	-0.6/-0.3	-5.5/+1.7	-14.5/+6.6	-17.5/+4.4	-1.7/+0.6	-45.0/+12.7	-0.8/0.0	-10.3/+2.0	-6.3/+1.1	-28.9/+18.7	-11.7/+5.1	-1.0/+0.7	-3.2/+2.8	-11.3/+4.3

[†] Native local features. Exact unit values and intervals are in Appendices E and F.

Taking the minimum directly implements the deployment principle motivated by the audit: complementary evidence is non-compensatory, so a single atypical channel is sufficient to veto acceptance.

The percentile transform is closely related to conformal ranking (Haroush et al. 2022; Bates et al. 2023; Kaur et al. 2022). Our contribution is not the ranking itself, but identifying the complementary evidence channels whose preservation improves deployment robustness. Because the channels are dependent and share the same calibration data, we make no finite-sample coverage guarantee.

Evaluation protocol. The primary evaluation uses five strict OpenOOD tasks with fixed unlabeled ID calibration data and held-out test sets. CEG is developed only on capped versions of these tasks, whereas the reported results use the full uncapped benchmarks. Transfer to SigLIP2 and PE-Core freezes all hyperparameters, and the protected operating point is evaluated using five-fold ID-only cross-fitting. Neither configuration uses OOD images or OOD labels during fitting. Complete protocols are given in Appendix B.

Deployment robustness. If the diagnosis is correct, preserving both channels should not only lower error but make the outcome less dependent on which detector was picked in the first place. That is the claim we test here, and it holds: CEG improves the family-balanced FPR95 of every wrapped detector, by between 9.3 and 35.9 points, and family AUROC rises for every base as well (Table 1), so the effect is not simply a threshold shift at 95% ID recall. The more interesting change, however, is not that each detector improves

but that they stop disagreeing. Among the five corpus-free bases, the spread between best and worst contracts from 29.9, 22.0 and 43.3 FPR95 points down to 3.3, 2.3 and 3.9, under family-balanced, unweighted 13-unit and MNIST-excluded weighting respectively (Appendix Table E6). Which guarded detector comes out ahead still shifts between these schemes, much as the audit would predict, but the contraction itself survives all three. This is what we mean by reducing sensitivity to the initial choice: the guard does not crown a winner, it makes the choice matter less. Table E6 therefore confirms the prediction of the portability audit: the largest corrections occur for detectors that previously discarded one complementary evidence channel. Accordingly, the strongest gains are observed for the one-cue detectors such as Energy and the gap-based scores, whereas Mahalanobis+CEG reaches the lowest overall error after its additional Gaussian fit and GL-MCM+CEG remains the strongest corpus-free endpoint.

A stricter benchmark could still be flattering the guard, so we ask whether the same interpretation holds across the broader audit, where 61 of the 65 CLIP detector-domain pairs improve. The remaining four cases are structured rather than scattered: they arise when one noisy channel vetoes an otherwise strong detector, which is precisely the trade-off imposed by a non-compensatory rule. The channel ablation supports the same interpretation (Appendix Table E3): level alone reaches 33.3 and sharpness alone 36.4 family-balanced FPR95, compared with 30.6 when both are preserved, while removing the local term costs a further 5.8 points. Simes and Fisher achieve marginally lower error on these development

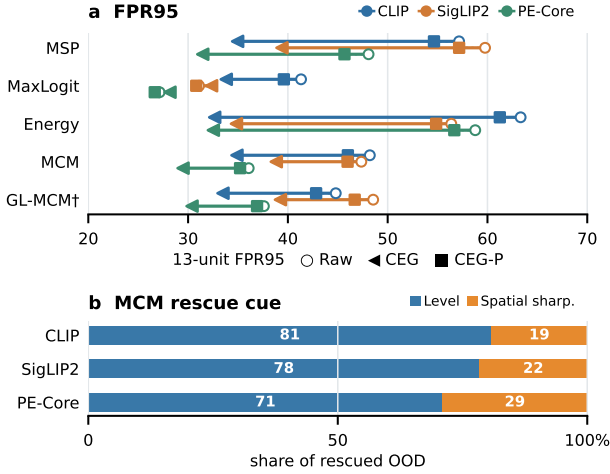


Figure 7: **CEG recovers complementary evidence.** (a) 13-unit FPR95 across five base detectors and three VLMs. (b) Most rescued MCM errors originate from the missing level cue rather than spatial sharpness.

tasks, but we retain the minimum because it is the only rule that directly implements the non-compensatory deployment principle motivated by the audit.

Does the evidence hypothesis explain the results? Figure 7 shows that the observed improvements follow the complementary evidence hypothesis. Panel (a) shows that CEG consistently produces the largest corrections for detectors that rely primarily on a single evidence channel, while the protected variant (CEG-P) retains most of those gains after reducing the strict veto behaviour. Rather than replacing existing detectors, the guard consistently improves them by recovering complementary evidence.

Panel (b) explains where these corrections originate. Among raw MCM errors rescued by CEG, the level channel provides 81/78/71% of the vetoes for CLIP/SigLIP2/PE, whereas spatial sharpness accounts for the remaining 19/22/29%. This is exactly what Proposition 1 predicts: because MCM removes absolute level, the dominant recovered evidence should be level. The remaining sharpness corrections show that local evidence contributes independently rather than merely duplicating the global signal. Together, these results close the loop between the mechanism and the deployment audit: different detectors discard different evidence channels, and preserving those complementary channels improves deployment robustness.

Channel-substitution controls. Adding any channel to a minimum increases conservatism, so we replace level and sharpness with entropy, logit variance, and seeded noise under identical calibration and evaluation (Figure 8). Entropy, variance, and noise all perform no better than raw MCM (47.2, 47.2, and 54.1 versus 42.6 FPR95), whereas sharpness, level, and both reduce FPR95 to 37.5, 32.0, and 30.5. The gain therefore follows the information contributed by the channel, not the conservatism of the minimum. Entropy is largely redundant with the sharpness evidence already

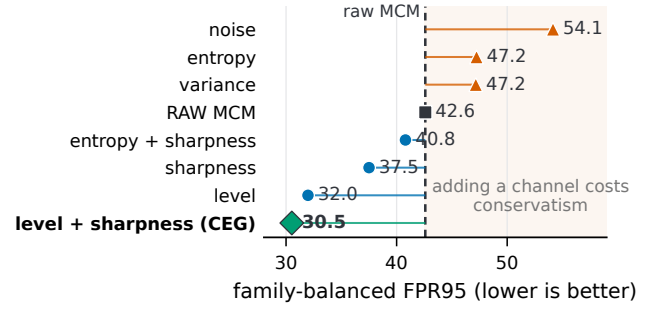


Figure 8: **Uninformative channels cost the minimum.** CLIP-B/16 MCM family-balanced FPR95 on five strict tasks and fixed ID-only manifests. Noise is seeded random; entropy and variance independently round to 47.2. Shading is worse than raw.

present in MCM, while replacing the best-anchor level signal with mean affinity improves only to 35.6, showing that the useful level evidence is carried by the best-anchor match rather than average affinity.

Broad deployment audit. We next ask whether the same principle holds across deployment settings. Evaluating CEG-P, a shrunk variant that retains two thirds of the base score, on five base detectors and three VLMs, far OOD improves in all 15 near/far deployment summaries and both near and far improve simultaneously in 14; every interval excludes zero. The sole exception is PE MaxLogit on near OOD (30.77 $\rightarrow$ 30.92), which still improves on the corresponding far benchmark (24.83 $\rightarrow$ 23.85). A symmetric fold-wise control that scores ID and OOD under identical protocols preserves all fifteen suite-level gains within 0.25 FPR95 points, so the improvements are not an artefact of the evaluation procedure. Full near/far tables, deployment-level regressions, and the near/far asymmetry analysis are in Appendix F.

6 Conclusion

We revisited a common assumption in zero-shot OOD detection: that the detector with the best benchmark performance is also the best deployment choice. A portability audit across ID domains and VLMs showed that this assumption does not hold. Detector rankings reverse because different ID domains preserve different evidence, with corpus-free methods relying primarily on level or sharpness and WordNet-based methods additionally depending on external semantic coverage. This diagnosis naturally motivates CEG, which preserves complementary evidence rather than committing to a single score, reducing deployment variability and tail risk without requiring OOD supervision, auxiliary corpora, or detector-specific adaptation. More broadly, our results suggest that evaluating zero-shot OOD detectors should move beyond benchmark averages toward deployment portability, explicit failure analysis, and revalidation whenever the ID domain or the underlying VLM changes.

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A Appendix Overview and Organization

This appendix provides the complete numerical evidence and implementation details behind the main paper. It keeps the failure audit, the benchmark-composition analysis, the strict CEG evaluation, and the later protected audit separate so that their resource assumptions and evidence status remain explicit. The material is organized as follows:

- **Section B: Detailed reproducibility protocol.** Frozen components, score and token definitions, datasets, calibration splits, resource accounting, uncertainty, aggregation, overlap policy, and fidelity checks. It also distinguishes stability, prospective, and development evidence.
- **Section C: Benchmark composition analysis.** Additional evidence motivating the use of OpenOOD-v1.5, including benchmark-overlap examples, nearest-neighbour analysis, and the comparison with the traditional ImageNet evaluation protocol.
- **Section D: Full failure audit.** Complete seventeen-domain CLIP results and Table B1, including the excluded-overlap row and the mean/worst-domain summaries underlying Figure 2.
- **Section E: Strict and prospective CEG evidence.** Suite uncertainty, the non-compensatory property of CEG, channel and combination ablations, architecture-transfer analysis, the disclosed nine-checkpoint robustness extension, and the pre-frozen ImageNet-10/ImageNet-20 confirmation.
- **Section F: Protected all-domain audit.** Near/far accounting, the symmetric cross-fit control, regression locations, complete Raw/CEG/CEG-P domain tables, and unaggregated ID $\times$ OOD-source results.

B Detailed Reproducibility Protocol

Frozen components and splits. All VLMs and score implementations are frozen. The strict CEG study uses five fixed manifests per task: 1,000 unlabeled train-ID images fit channel standardisers and empirical CDFs, a disjoint 1,000 train-ID images set the deployment operating threshold, and official test images remain evaluation only. The CLIP-to-SigLIP2/PE port freezes tasks, prompts, channel forms, hyperparameters, and full-image evaluation before reading transfer OOD results. No experiment fits target-OOD images.

Evidence tiers. The channel forms and $k=10$, $Q_{0.9}$, 3×3 kernel, and minimum fusion were developed on capped versions of the same five strict task domains; their uncapped rerun is sampling-level stability evidence, not new-task confirmation. The multi-VLM port and reciprocal ImageNet-10/20 experiment are prospective. CEG-P was defined after observing transfer failures and its 18-dataset cross-fit is explicitly post-result development evidence.

Resource and uncertainty accounting. Corpus-free global scores use only ID class names; GL-MCM additionally uses native local tokens, Mahalanobis fits extra unlabeled-ID feature statistics, and NegLabel/CSP use WordNet. DomainNet counts once in aggregates. Reported intervals use 5,000 paired image/calibration resamples; broad

cellwise intervals are nominal and unadjusted. The details below give the exact score equations, datasets, folds, aggregation, overlap policy, and fidelity checks.

Scores and VLM features. ID anchors use the dataset class names with the template a photo of a $\{\}$.; NegLabel keeps its original the nice $\{\}$ template and published NegMining pipeline over roughly 10k WordNet negatives. CSP uses the published Conjugated Semantic Pool: WordNet noun/adjective prompt ensembles are mined against the ID vocabulary, then grouped to score positive softmax mass. Our deterministic replay sorts corpus filenames before seeded adjective conjugation. For normalised image, patch, and text embeddings g, p_i, t_c , we write $\ell_c = \langle g, t_c \rangle$ and $\ell_{i,c} = \langle p_i, t_c \rangle$. The audit implements MSP at the native logit scale (Hendrycks and Gimpel 2017), MaxLogit (Hendrycks et al. 2022), Energy (Liu et al. 2020), MCM (Ming et al. 2022), GL-MCM (Miyai et al. 2025), and NegLabel (Jiang et al. 2024). Mahalanobis fits class means and a shared covariance to zero-shot ID pseudo-labels, never ground-truth labels (Lee et al. 2018). CLIP local scores use projected final-block MaskCLIP-style value features (Zhou, Loy, and Dai 2022; M. De la Jara et al. 2025). The frozen transfer port uses SigLIP2’s 32×32 post-normalisation patch sequence before pooling and PE’s projected 14×14 final sequence without its class token. Because these native token paths are not representation-identical, MCM is the architecture-neutral transfer base; GL-MCM is a native-token diagnostic rather than a representation-equivalent port.

The strict scalar cache predates the multi-VLM audit and lacks the native-logit scale MSP channel, so MCM is its softmax baseline; MSP is reported throughout the broad audit where it was frozen. The broad matched set comprises the same five defined scalar bases for every dataset and VLM at the audit freeze (MSP, MaxLogit, Energy, MCM, and GL-MCM), not a result-selected subset. Figure 2 keeps the evidence tiers explicit: the four-score global comparison was frozen before the uncapped rerun, whereas GL-MCM is shown separately as a post-result native-token extension over the complete cache.

For Table 1, Mahalanobis is fitted on 1,000 unlabeled train-ID images deterministically held out from every guard fit and operating manifest. This prevents the empirical guard CDF from calibrating on the Gaussian’s own training scores while making the extra ID-fitting resource explicit.

Datasets and OOD pools. The CLIP mechanism audit contains ImageNet-1K/200; CUB-200, Food-101, Oxford Pets, and Caltech-101; EuroSAT, DTD, and Pascal VOC; five non-photographic DomainNet styles; and CIFAR-10/100 and MNIST (Deng et al. 2009; Wah et al. 2011; Bossard, Guillaumin, and Van Gool 2014; Parkhi et al. 2012; Fei-Fei, Fergus, and Perona 2004; Helber et al. 2019; Cimpoi et al. 2014; Everingham et al. 2010; Peng et al. 2019; Krizhevsky 2009; LeCun et al. 1998). The uncapped audits also include DomainNet Real, yielding 18 raw datasets and 13 display units after its six styles are averaged once. The historical table uses fixed 2,400-ID/1,500-OOD draws from the five-source ImageNet OpenOOD-v1.5 pool: SSB-hard, NINCO, iNaturalist, Textures, and OpenImage-O (Zhang et al. 2023; Vaze et al.

Table B1: **Seventeen-domain CLIP audit**. FPR95 $\downarrow$ (all-OOD mean, %). Green cells mark the row winner; the header bands expose resource assumptions. Every detector exceeds 80 somewhere. DTD † is excluded from Mean/Worst because its OOD suite contains DTD-derived textures.

Family	ID dataset	Global class-name scores				Native local	ID fit	WordNet
		MSP	MaxLogit	Energy	MCM	GL-MCM	Mahalanobis	NegLabel
ImageNet	ImageNet-1K	68	75	97	61	51	96	47
	ImageNet-200	36	47	95	34	27	84	32
Fine-grained	CUB-200	77	2	21	23	47	2	2
	Food-101	23	6	50	10	6	3	3
	Oxford Pets	34	0	22	8	11	0	0
	Caltech-101	34	34	95	27	22	27	24
Far domains	EuroSAT	98	61	22	98	88	0	38
	DTD †	86	76	83	87	73	38	51
	Pascal VOC	57	58	93	52	38	27	20
Renderings	Quickdraw	97	40	16	97	96	0	45
	Clipart	82	97	100	85	82	37	37
	Painting	89	99	99	90	83	90	50
	Sketch	88	97	99	88	83	56	42
	Infograph	90	100	100	87	86	81	49
Small/coarse	CIFAR-10	24	15	70	18	18	17	36
	CIFAR-100	87	65	78	83	68	39	89
	MNIST	35	24	33	32	32	0	70
Mean (16 rows)		63.7	51.3	68.1	55.8	52.4	34.9	36.5
Worst (16 rows)		98	100	100	98	96	96	89

2022; Bitterwolf, Müller, and Hein 2023; Van Horn et al. 2018; Wang et al. 2022). Its DTD-as-ID row is excluded from Mean/Worst because that pool contains DTD-derived OOD images. The uncapped audits omit Textures from all shared ImageNet reference probes, removing this self-overlap; CIFAR and MNIST retain their native near/far suites. Shared probes hold OOD inputs fixed while ID changes. Beyond removing exact DTD/Textures self-overlap, we do not filter semantic class overlap because the shared probe sources lack a complete label mapping to each ID vocabulary. These rows permit category overlap by design and are reference-shift stress tests, not strict semantic-novelty benchmarks. Rendering rows likewise test category plus rendition shift.

Primary guard and prospective transfer. The strict suite is ImageNet-1K, ImageNet-200, CIFAR-10, CIFAR-100, and MNIST. The level/sharpness forms, $k=10$, $Q_{0.9}$, 3×3 kernel, and minimum fusion were developed on capped versions of these same five task domains. Freezing them before the uncapped rerun prevents image-level leakage but does not make the strict headline task-level confirmatory; only the reciprocal ImageNet-10/20 check and prospective VLM port are out-of-development tests. For each task, five fixed manifests use 1,000 unlabeled train-ID images to fit channel standardisers/CDFs and a disjoint 1,000 train-ID images for the deployment operating threshold; all official test images are reserved for evaluation. The reported benchmark FPR95 convention instead sets the 95%-recall threshold on evaluation ID. The primary aggregate first averages within ImageNet and CIFAR and then weights ImageNet, CIFAR, and MNIST equally. Before reading SigLIP2/PE OOD results, we froze the tasks, manifests, prompt, guard hyperparameters, and full-image policy; no architecture-specific layer, token, task,

projection, or weight sweep was performed. Paired replicates share physical-image and calibration-manifest draws across VLMs.

Post-result protected audit. The broader CEG-P study is deliberately labelled development evidence. For datasets without a designated calibration split, and uniformly throughout this audit, we use five-seed, five-fold ID-only cross-fitting. In each fold a guard is fitted on a deterministic sample of up to 1,000 images from the other four folds; if fewer are available, it uses every non-held-out ID image. Thus every ID image receives a score from a guard that did not fit that image. For example, VOC-13 has 960 images, so each guard fits approximately 768 and scores the remaining approximately 192. Each OOD image receives the mean score over the five guards, and no OOD image enters guard fitting. ID labels only stratify fold assignment. Thus this cross-fit can use more unique ID images than the primary calibration and applies different ensemble constructions to ID and OOD. The same paired physical-image and cross-fit draws are shared across scores, tasks, variants, and VLMs; external-corpus scores are excluded from this matched comparison. The symmetric fold-wise control in Appendix F removes this ID/OOD scoring asymmetry. Mahalanobis is also excluded: unlike the common cached score channels, it requires a backbone-specific ID-feature fit, and equivalent global-feature caches were not retained for all three VLMs. CSP is excluded because its external mined prompt pool is not resource-matched across the three VLMs.

Sanity checks. On the classic iNaturalist/SUN/Places/Textures ImageNet protocol (Xiao et al. 2010; Zhou et al. 2017), our MCM and NegLabel imple-

mentations obtain 42.7 and 26.0 FPR95, respectively, versus published 43.9 and 25.4. Zero-shot top-1 accuracy on the strict suite is 67.09, 91.77, 89.73, 66.70, and 38.43% in task order. The low MNIST accuracy is a caveat: better rejection alone does not make the classifier deployment-ready.

C Why We Use OpenOOD v1.5

The benchmark label does not guarantee sample-level separation. The traditional zero-shot ImageNet protocol established by MOS (Huang and Li 2021) uses iNaturalist, SUN, Places, and Textures as four far-OOD sources. Figure C1a shows selected high-similarity cross-dataset pairs. Three illustrate shared visual domains rather than proven leakage. The Places case is qualitatively different: `i_igloo_00004659.jpg` and ImageNet validation image `ILSVRC2012_val_00042483.JPEG` contain the same aligned 600×480 photograph. Their pixels differ only at encoding level (PSNR 51.5 dB; 72.9% of RGB channel values are exactly equal), making this a verified cross-split duplicate rather than merely a semantic neighbour.

A matched nearest-neighbour diagnostic. To test whether visual proximity extends beyond hand-picked examples, we stratify ImageNet validation within each of its 1,000 classes (seed 0): 25 images per class form a 25k gallery and the remaining 25 form a held-out ID query set. For each held-out ID image and 5k images from each traditional OOD source, we compute its maximum cosine similarity to the same gallery using frozen CLIP ViT-B/16 image embeddings. At the threshold retaining 95% of held-out ID, 93.7–99.3% of these nominally OOD images also pass (Figure C1b). This one-nearest-neighbour probe is a visual-domain diagnostic, not an estimate of duplicate prevalence and not a claim that the datasets are impossible to distinguish. It shows that visual proximity alone does not cleanly realize the intended far-OOD boundary.

C.1 Audit Protocol

Models and detectors. The audit separates mechanism from portability. We therefore study the evidence mechanism using frozen CLIP ViT-B/16 (Radford et al. 2021), then replicate the conclusions on SigLIP2-B/16 (Tschannen et al. 2025) and PE-Core-B/16 (Bolya et al. 2025) with their native preprocessing. The comparison includes MSP, MaxLogit, Energy, Mahalanobis, MCM, GL-MCM and NegLabel, before later wrapping ten structurally different detectors with the same guard. ID class names provide the text anchors, while NegLabel retains its published WordNet pipeline. Mahalanobis requires an additional ID-feature fit and is therefore reported as a separate resource tier. Exact score definitions are given in Appendix B.

Deployment settings. The audit combines a 17-dataset CLIP mechanism study with an uncapped 18-dataset, three-VLM replication spanning five deployment families. Uncapped experiments retain every evaluation image and each dataset’s native vocabulary; CIFAR-10/100 and MNIST use their native OpenOOD suites, while the shared ImageNet reference pool is fixed before scoring. This design isolates de-

tector portability across both deployment domains and model families.

Evaluation criteria. Detector portability is measured using FPR95↓ at 95% ID recall, averaged across each suite’s OOD splits. DomainNet contributes once to cross-domain aggregates. Unless stated otherwise, all experiments are fully zero-shot and use frozen models, without fitting to OOD images or ground-truth OOD labels. Complete implementation details are given in Appendix B.

D Full Seventeen-Domain Audit

How to read the audit. The mean and worst rows answer different deployment questions. Mahalanobis has the lowest cross-domain mean, but its ImageNet failures leave a 96 worst case; NegLabel has the best worst case, yet fails badly on CIFAR-100 and MNIST. The resource bands make this trade-off explicit: an ID-feature fit or external corpus can move the failure rather than remove it. The within-family reversals also rule out a simple “photographic versus non-photographic” explanation; even related fine-grained datasets reward different scores.

Evidence boundary. Table B1 contains raw detector scores only; no row uses CEG or CEG-P. DTD is shown for transparency but excluded from Mean/Worst because its declared OOD pool contains DTD-derived textures. Section E next evaluates the fixed minimum guard, while Section F keeps the later protected development study separate.

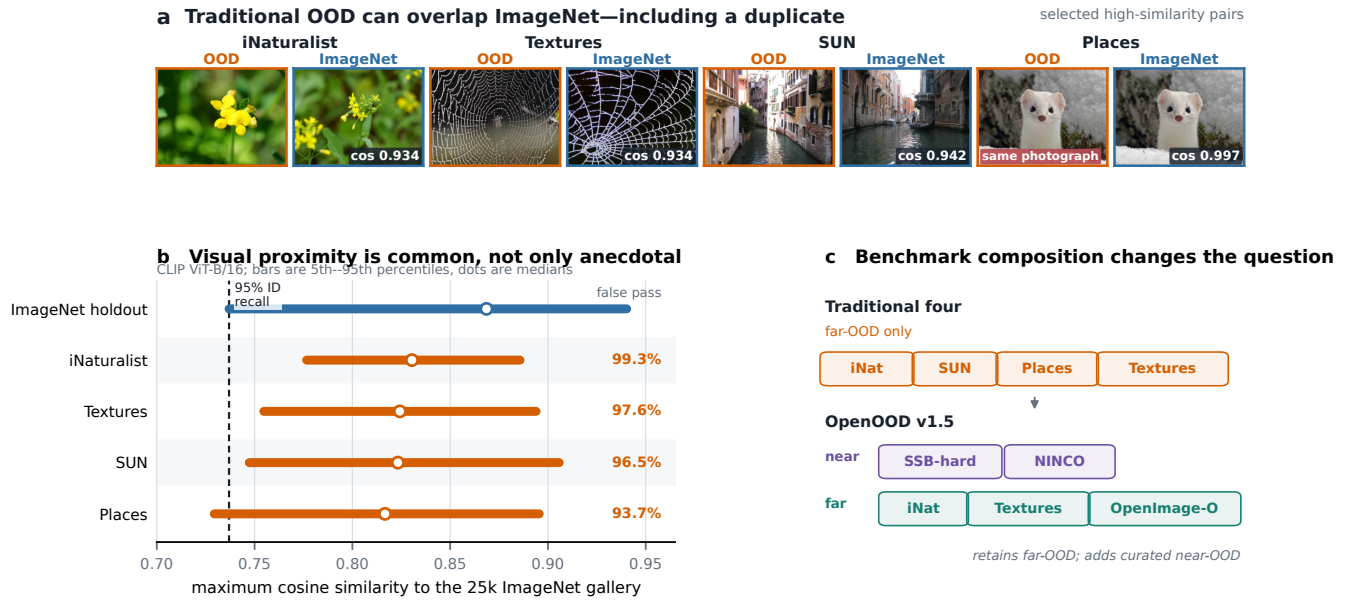


Figure C1: **One label, a wide range of proximity.** (a) Pairs from the traditional four-source ImageNet suite, ordered by similarity: all are labelled far-OOD, yet they range from shared visual domain to a verified re-encoded duplicate of an ImageNet validation image. (b) Maximum cosine similarity to a matched 25k ImageNet gallery. Bars span the 5th–95th percentiles and circles mark medians; the dashed line is the threshold retaining 95% of the held-out ImageNet queries. Orange percentages are the OOD false-pass rates at that threshold. (c) The traditional protocol exposes only four far-OOD sources. OpenOOD-v1.5 retains iNaturalist and Textures, adds OpenImage-O, and separately exposes the curated near-OOD SSB-hard and NINCO sources.

E CEG Evidence and Uncertainty

E.1 Non-compensatory property of CEG

Equation (2) defines the guard as

$$G_B(x) = \min_{R \in \{B, L, S\}} U_R(x).$$

The minimum has the following elementary property.

Proposition 2. For every $\tau \in [0, 1]$,

$$\{G_B \geq \tau\} = \bigcap_{R \in \{B, L, S\}} \{U_R \geq \tau\},$$

and therefore, for any test distribution Q ,

$$Q(G_B \geq \tau) \leq \min_R Q(U_R \geq \tau).$$

Proof. A minimum is at least τ if and only if every argument is at least τ . The probability inequality follows immediately because the intersection is contained in each individual event. $\square$

This proposition formalises the non-compensatory behaviour of CEG. It explains why a single atypical evidence channel can veto acceptance, but it does not imply improved OOD performance after calibration. The empirical benefits of this design are evaluated in Section 5.

Image-level intuition. Figure E1 illustrates the non-compensatory behaviour formalised by Proposition 2. The texture matches the ID classes strongly in absolute level but lacks a coherent class peak, whereas the street scene has a sharp “truck” winner despite an atypically weak absolute match. Because the missing cue changes across samples, preserving both channels allows either one to veto acceptance when it becomes atypical.

Table E1: **Suite-level uncertainty for CEG.** Δ is CEG minus Raw family-balanced FPR95; every interval is below zero and is marked green. The intervals use 5,000 paired image/calibration resamples and summarize the whole five-task suite, not an individual domain.

Base	Δ FPR95	Paired 95% interval
MaxLogit	-10.16	[-10.95, -9.56]
Energy ($\tau=1$)	-35.93	[-36.86, -35.08]
MCM	-12.07	[-12.99, -11.49]
GL-MCM [†]	-9.28	[-9.92, -8.73]
20%-nonmax gap (abl.)	-28.93	[-30.09, -27.99]
LogitGap	-19.39	[-20.28, -18.44]
Mahalanobis	-18.22	[-18.83, -17.80]
NegLabel	-27.20	[-28.23, -25.90]
CSP	-21.17	[-22.08, -20.10]

Protected frontier across native OpenOOD. Table E2 reports the deployment choice on the four native non-digit tasks while retaining MNIST in the complete five-task analyses. Every displayed nonraw point improves the mean for all seven bases. This includes the external-corpus detectors: NegLabel

Base detector	Raw	CEG-P ($\lambda=1/3$)	$\lambda=0.50$	$\lambda=0.75$	Hard CEG
Class-name scores; no external corpus					
MaxLogit	48.58	45.47	43.53	39.76	36.92
Energy ($\tau=1$)	84.79	83.20	82.03	78.04	39.95
MCM	48.49	45.67	44.25	41.37	37.50
GL-MCM [†]	41.50	40.09	39.22	37.59	35.99
20%-nonmax gap [‡]	52.73	48.75	46.65	42.37	37.59
External WordNet corpus/mining					
NegLabel	50.08	47.61	45.95	42.88	35.62
CSP-deterministic	44.49	41.66	40.34	38.15	33.16

Table E2: **Protected CEG deployment frontier on native OpenOOD, excluding MNIST.** All entries are mean FPR95 $\downarrow$ with CLIP-B/16. OOD sources are first averaged within each task, after which ImageNet-1K, ImageNet-200, CIFAR-10, and CIFAR-100 receive equal weight. This is a post-result sensitivity view: MNIST remains in the complete five-task analyses but is excluded here so that its qualitatively distinct coarse digit regime does not dominate this mean. The declared CEG-P point uses $\lambda = 1/3$; $\lambda = 0.50$ and 0.75 are additional fixed post-result deployment points, and $\lambda = 1$ is Hard CEG. No point is selected from these OOD results. Every displayed nonraw point improves the four-task mean for all seven bases.

[†] Uses native local features. [‡] Represents the numerically equivalent fixed- N LogitGap family in this protection sweep.

moves from 50.08 to 47.61, 45.95, 42.88, and 35.62 as λ increases, while CSP moves from 44.49 to 41.66, 40.34, 38.15, and 33.16. Thus the aggregate conclusion does not rely on omitting strong external-semantic baselines. The frontier is descriptive post-result evidence, not an OOD-selected operating point.

Table E3: **Channel and combination ablation.** Family-balanced FPR95 $\downarrow$ on the five strict CLIP tasks. Green/red denotes lower/higher FPR95 than the gray Raw reference by more than 0.10 points; gray denotes a practical tie. “Global” removes local-token cues; the last column averages all five corpus-free bases. This is same-task development evidence.

Rule	MCM	MaxL.	Energy	GL	All
Raw base	42.6	40.2	68.0	38.1	49.9
Base $\wedge$ MaxLogit	35.8	40.7	47.8	31.6	39.4
Base $\wedge$ level only	32.0	33.7	37.1	29.1	33.3
Base $\wedge$ sharpness only	37.5	31.2	35.6	35.7	36.4
Global-only guard	35.8	35.8	40.1	32.2	36.4
Full CEG (minimum)	30.5	30.1	32.1	28.8	30.6
Mean of three ranks	32.6	31.9	47.7	31.6	35.3
Simes	29.9	29.1	31.2	28.2	29.9
Fisher	30.0	29.1	29.9	28.6	29.6

Mean gain is bought with tail risk. Table E4 sweeps the shrinkage weight λ from the raw base to the hard minimum and beyond. Within the convex regime the two columns move together: larger mean corrections come with larger worst-case regressions, from $-0.41/+0.43$ at $\lambda = 0.1$ to $-5.71/+16.04$ at the hard guard, which also regresses on 55 of 135 task cells. CEG-P ($\lambda = 1/3$) sits where the trade is most favourable: it improves all 27 checkpoint aggregates while bounding the worst task regression at $+1.01$. Past $\lambda = 1$ the score is no longer a convex combination, and vetoing more aggressively fails outright ($+52.16$ mean, $+96.86$ worst at $\lambda = 2$). Reliability therefore does not increase monotonically with conservatism; as in the channel substitutions of Figure 8, it depends on how much evidence

λ	Regime	Mean Δ	Worst Δ	Task I/T/W	Agg. I/T/W
0	Raw base	+0.00	+0.00	0/135/0	0/27/0
0.1	Interpolation	-0.41	+0.43	95/29/11	26/1/0
0.2	Interpolation	-0.82	+0.56	102/19/14	27/0/0
1/3	CEG-P	-1.47	+1.01	110/11/14	27/0/0
0.5	Interpolation	-2.34	+2.16	112/6/17	27/0/0
0.75	Interpolation	-4.16	+5.91	103/8/24	27/0/0
1	Hard CEG	-5.71	+16.04	76/4/55	22/0/5
1.25	Amplification	+23.89	+96.65	26/0/109	3/0/24
1.5	Amplification	+48.56	+96.81	9/0/126	0/0/27
2	Amplification	+52.16	+96.86	3/0/132	0/0/27

Table E4: **CEG deployment risk-reward frontier.** $S_\lambda = U_B - \lambda(U_B - \min\{U_B, U_L, U_S\})$ over 3 bases, 9 checkpoints, and 5 strict tasks. Δ is wrapped minus raw FPR95, so negative is better. I/T/W counts improvements, practical ties ($|\Delta| \leq 0.10$), and regressions. “Worst” is the largest task-level regression. Green/red/gray denotes improvement/regression/practical tie to Raw in the two Δ columns. The complete fixed grid is post-result descriptive evidence; no λ is selected from these OOD results. $\lambda = 1/3$ and 1 are the declared CEG-P and hard-CEG references. Amplification ($\lambda > 1$) is outside the protected convex regime and fails sharply.

the added veto actually carries.

Capacity-matched combination control. Table E3 gives every combination rule exactly the same three ranks U_B, U_L, U_S . Their compensatory mean reaches 35.3 FPR95, versus 30.6 for the minimum, so the gain is not explained merely by exposing the detector to three channels. Simes and Fisher reach 29.9 and 29.6 on these same development tasks, so their numerical advantage is real. It answers a different design objective: Fisher can offset one atypical rank with typical ranks, while Simes discounts an isolated atypical rank unless multiple channels agree. Both relax the single-channel veto and can therefore suppress the missing-cue alarm iden-

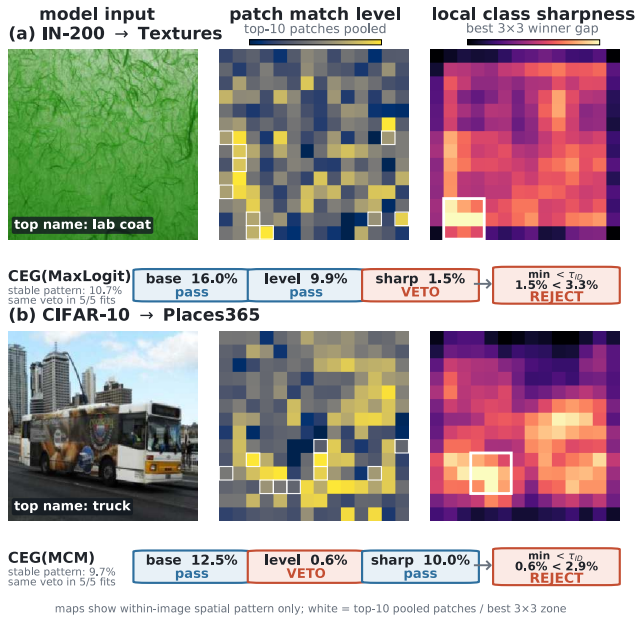


Figure E1: **Image-level non-compensation in CEG.** Representative conditional-median examples show opposite vetoes. A texture passes the base detector and level but fails sharpness, whereas a bus passes the base detector and sharpness but fails level. These patterns are stable across all five frozen ID-only fits.

tified by the audit. We retain the minimum as the declared non-compensatory rule, not as the post-hoc numerical optimum.

Tie-margin sensitivity. In the nine-checkpoint deployment frontier, at $\lambda = 1/3$, task I/T/W counts are 114/4/17, 110/11/14, and 105/22/8 for margins 0.05, 0.10, and 0.20, respectively. All 27/27 checkpoint/method aggregates improve at every margin; the same 27/27 aggregate conclusion holds for $\lambda = 0.50$ and 0.75.

Protected-weight provenance. The 2:1 ratio was chosen after inspecting the transfer failures it was designed to mitigate; it is not a held-out hyperparameter choice, and no CEG-P comparison is presented as confirmation. Every CEG-P table and the fixed λ frontier are explicitly post-result development evidence. The score uses no OOD sample during fitting or inference, but that resource property does not make the weight selection prospective.

Prospective transfer details. For frozen MCM+CEG, the paired family-macro changes are -12.07 $[-12.99, -11.48]$ for CLIP, -14.27 $[-15.49, -13.38]$ for SigLIP2, and -8.60 $[-9.11, -7.85]$ for PE. Both SigLIP2 ImageNet tasks worsen, as do PE ImageNet-200, CIFAR-10, and CIFAR-100: five of 15 point estimates. Four paired intervals lie above zero and PE CIFAR-100 remains unresolved. CEG(MaxLogit) also worsens the SigLIP2 aggregate from 28.77 to 30.25.

Post-result checkpoint extension. Table E5 crosses three CLIP, three SigLIP2, and three PE-Core checkpoints with

every detector base available throughout the strict cache: MCM, MaxLogit and GL-MCM. Tasks, images, calibration manifests, prompts, and guard hyperparameters remain fixed. CEG-P improves all 27 detector/checkpoint aggregates and 110/135 task cells; its nine detector/family mean changes range from -0.79 to -2.55 . Hard CEG produces larger mean gains but improves only 22/27 aggregates and 76/135 task cells. Its clearest failure is MaxLogit on SigLIP2, whose family mean worsens by 1.54. Thus protection generalizes beyond MCM and better tolerates architecture-dependent channel failures. Because the six additional checkpoints were chosen after the original transfer was read, the extension is robustness evidence rather than prospective confirmation.

Pre-frozen near-OOD confirmation. Under the pre-frozen reciprocal ImageNet-10/20 protocol, seven of eight intervals exclude zero: MCM, GL-MCM, MaxLogit, Energy, fixed-gap, LogitGap, and NegLabel. CSP $[-1.75, 1.25]$ is unresolved. Because these are correlated ImageNet subsets, the multi-family suite remains primary.

Table E5: **CEG-P and Hard CEG across three detector bases and nine checkpoints.** Each family row equally averages its three checkpoints; every checkpoint value equally averages the five strict tasks after within-task OOD-source aggregation. All numeric scores are FPR95↓, and Δ is wrapped minus Raw, so negative is better. I/T/W gives improvements, practical ties, and regressions; counts before and after the semicolon refer to checkpoint aggregates and task cells. CEG-P improves 27/27 detector/checkpoint aggregates, whereas Hard CEG improves 22 and worsens 5. The evaluated bases are MCM, MaxLogit, GL-MCM[†]. Energy is absent because its native-scale score was not retained for all six added checkpoints, although it is available for the original CLIP-B/16 strict evaluation in Table 1; NegLabel and CSP are absent because their external-corpus scores were not recomputed for those checkpoints. Methods are unavailable rather than result-filtered. The extension is post-result robustness evidence.

VLM family	Base detector B	Raw	CEG-P	Δ_P	P I/T/W (agg.; task)	Hard CEG	Δ_H	H I/T/W (agg.; task)
CLIP								
CLIP	MCM	45.83	43.27	-2.55	3/0/0; 14/1/0	32.91	-12.92	3/0/0; 11/0/4
CLIP	MaxLogit	38.78	36.89	-1.89	3/0/0; 15/0/0	31.95	-6.83	3/0/0; 12/0/3
CLIP	GL-MCM [†]	45.76	43.48	-2.28	3/0/0; 14/0/1	32.52	-13.24	3/0/0; 11/0/4
SigLIP2								
SigLIP2	MCM	59.61	58.09	-1.52	3/0/0; 10/4/1	47.37	-12.24	3/0/0; 9/0/6
SigLIP2	MaxLogit	32.31	31.52	-0.79	3/0/0; 12/1/2	33.85	+1.54	0/0/3; 5/1/9
SigLIP2	GL-MCM [†]	61.72	59.71	-2.01	3/0/0; 12/2/1	49.20	-12.53	3/0/0; 9/0/6
PE-Core								
PE-Core	MCM	38.82	37.38	-1.44	3/0/0; 12/1/2	33.70	-5.12	2/0/1; 6/0/9
PE-Core	MaxLogit	29.93	28.94	-0.99	3/0/0; 11/1/3	28.75	-1.18	2/0/1; 8/1/6
PE-Core	GL-MCM [†]	41.41	40.44	-0.97	3/0/0; 10/1/4	36.14	-5.27	3/0/0; 5/2/8
All nine checkpoints								
All	MCM	48.09	46.25	-1.84	9/0/0; 36/6/3	37.99	-10.09	8/0/1; 26/0/19
All	MaxLogit	33.67	32.45	-1.22	9/0/0; 38/2/5	31.52	-2.16	5/0/4; 25/2/18
All	GL-MCM [†]	49.63	47.88	-1.75	9/0/0; 36/3/6	39.29	-10.34	9/0/0; 25/2/18

[†] GL-MCM uses native local tokens. Practical ties satisfy $|\Delta| \leq 0.10$ and are descriptive rather than statistical equivalence.

Table E6: **Aggregation robustness across evaluation weightings.** Each cell reports Raw→CEG FPR95↓/AUROC↑ for five base detectors. For MaxLogit, Energy, MCM, and GL-MCM, Fam. and no-MN use the final strict protocol: Fam. balances ImageNet, CIFAR, and MNIST, while no-MN excludes MNIST and rebalances ImageNet and CIFAR. The 13-unit column instead uses the independently frozen all-image broad cache of Tables F2–F4, with every unit weighted equally and DomainNet counted once. MSP is absent from the strict cache, so all three MSP cells use the broad all-image protocol. Comparisons should be made within a column.

Base	Fam.	13-unit	no-MN
MSP	48.4→30.7/86.3→91.8	57.2→35.0/78.5→88.8	53.9→38.0/84.3→89.3
MaxLogit	40.2→30.1/89.8→92.2	41.3→33.8/86.5→89.2	48.6→37.2/87.1→89.8
Energy	68.0→32.1/67.5→91.3	63.3→32.7/69.8→90.0	84.8→40.1/56.0→88.4
MCM	42.6→30.5/88.7→92.1	48.2→34.9/81.7→89.0	48.5→37.8/86.3→89.6
GL-MCM [†]	38.1→28.8/89.4→92.3	44.8→33.5/85.0→89.3	41.5→36.2/87.6→89.7

[†] Native local features; the comparison matches the five-base main-paper scope.

F Protected All-Domain Audit

Broad failure locations. Across all five disclosed bases, nominal unadjusted intervals classify 136 display units as improvements, 49 as inconclusive, and 10 as regressions. Regressions concentrate in strong domain-specific bases, notably Energy/MaxLogit on EuroSAT and MSP/MCM on DomainNet. For MCM alone, paired suite changes are -2.19 $[-2.54, -1.83]$ for CLIP, -1.43 $[-1.86, -1.02]$ for SigLIP2, and -0.92 $[-1.33, -0.54]$ for PE; its sole resolved regression is PE DomainNet ($89.43 \rightarrow 89.74$).

Symmetric cross-fit control. The original broad protocol scores held-out ID with one fold-specific guard but averages five guard scores for each OOD image. We recomputed every point estimate foldwise, scoring held-out ID and OOD with the same guard before averaging folds. All 15 suite deltas remain negative; relative to the reported construction, changes lie between -0.21 and 0.00 FPR95 points. The asymmetry is therefore real and disclosed, but does not explain the broad mean gains.

Trade-offs and alternatives. The hard guard is not a Pareto fix: Table E4 shows a $+16.04$ worst task regression and 55/135 regressed task cells at $\lambda = 1$, whereas CEG-P limits the worst regression to $+1.01$ and improves 110/135 cells. Regressed residual fusion, spatial coherence, and weighted addition hurt strong bases or traded mean against worst case. Simes/Fisher modestly beat the minimum on the development suite (Table E3) but lose its strict inter-section semantics and were not prospectively transferred.

Table F1: **Near- and far-ODD point macros for the broad protected audit.** Entries are Raw $\rightarrow$ CEG-P FPR95 $\downarrow$; Δ is wrapped minus Raw. Gray marks a practical tie, $|\Delta| \leq 0.10$. Each value equally averages the 13 display units (DomainNet counts once). Full-suite paired intervals are in Table E1.

VLM	Base	Near OOD		Far OOD	
		Raw $\rightarrow$ CEG-P	Δ	Raw $\rightarrow$ CEG-P	Δ
CLIP-B/16	MSP	65.61 $\rightarrow$ 63.58	-2.03	49.76 $\rightarrow$ 46.93	-2.83
	MaxLogit	43.24 $\rightarrow$ 42.32	-0.92	39.90 $\rightarrow$ 37.56	-2.33
	Energy	58.73 $\rightarrow$ 56.63	-2.10	67.38 $\rightarrow$ 65.26	-2.13
	MCM	58.39 $\rightarrow$ 56.29	-2.11	39.35 $\rightarrow$ 37.22	-2.13
	GL-MCM [†]	54.93 $\rightarrow$ 52.88	-2.04	36.17 $\rightarrow$ 34.35	-1.82
SigLIP2-B/16	MSP	62.95 $\rightarrow$ 60.65	-2.31	56.61 $\rightarrow$ 53.83	-2.78
	MaxLogit	36.59 $\rightarrow$ 36.40	-0.19	27.42 $\rightarrow$ 27.09	-0.33
	Energy	59.71 $\rightarrow$ 58.27	-1.45	55.01 $\rightarrow$ 53.40	-1.62
	MCM	52.86 $\rightarrow$ 51.44	-1.42	42.55 $\rightarrow$ 41.23	-1.32
	GL-MCM [†]	54.48 $\rightarrow$ 52.77	-1.71	43.43 $\rightarrow$ 41.58	-1.85
PE-Core-B/16	MSP	57.65 $\rightarrow$ 55.40	-2.25	40.67 $\rightarrow$ 38.17	-2.50
	MaxLogit	30.77 $\rightarrow$ 30.92	$+0.16$	24.83 $\rightarrow$ 23.85	-0.98
	Energy	55.58 $\rightarrow$ 53.71	-1.86	62.86 $\rightarrow$ 60.47	-2.40
	MCM	46.53 $\rightarrow$ 45.70	-0.83	28.02 $\rightarrow$ 27.17	-0.85
	GL-MCM [†]	48.70 $\rightarrow$ 47.92	-0.78	28.94 $\rightarrow$ 28.38	-0.56

[†] Uses native local tokens.

CEG-P is the disclosed post-result compromise: it shrinks CEG toward the base, improving transfer stability while giving up much of the minimum rule’s aggregate gain.

Complete domain-level results. Tables F2 to F4 report FPR95 and AUROC for Raw, CEG, and CEG-P on every one of the 13 display units and each VLM. They use all evaluation images and preserve the post-result status of this broad audit; the tables are descriptive rather than an independent confirmation.

Unaggregated source transparency. Tables F5 to F9 expand the MCM base used by Figure 7 into every raw ID dataset and every declared OOD source. Unlike the display-unit tables, it does not average DomainNet styles or near/far sources. The complete machine-readable source table additionally includes all five bases, three variants, both FPR95 and AUROC, and all three VLMs.

At this source resolution, CEG-P improves 896 of 1,185 base/VLM/ID/ODD point comparisons, leaves 51 unchanged, and worsens 238; far sources improve more often and more strongly (494/645, -1.40 points on average) than near sources (402/540, -1.05). For MCM alone the counts are 183/237 improved, six unchanged, and 48 worsened. The largest protected gains are CLIP CIFAR-100 $\rightarrow$ MNIST (-7.6) and MNIST $\rightarrow$ Places365 (-7.1), while the recurring failure is DomainNet-real $\rightarrow$ iNaturalist ($+2.1$ to $+3.0$) on every VLM. This natural-to-natural failure is consistent with the guard’s boundary: if OOD remains ID-like in the base, level, and sharpness channels, there is no missing cue for a minimum to veto. Source averaging would hide this structured failure as apparently random noise.

Table F2: **Full all-domain CEG audit: CLIP-B/16**. Each cell is FPR95↓/AUROC↑ on every image; Raw is the base score, CEG is the non-compensatory minimum, and CEG-P is the protected blend. For guarded rows, green/red denotes FPR95 lower/higher than Raw by more than 0.10 points and gray denotes a practical tie; color is keyed to FPR95 only. Values average OOD sources within a task; DomainNet averages its six styles once. This is post-result development evidence.

Native and natural/fine-grained units								
Base	Variant	IN-1K	C-10	C-100	MNIST	IN-200	CUB	Food
MSP	Raw	66.4/78.1	26.0/93.7	86.9/75.1	37.3/90.1	36.3/90.3	79.0/68.1	20.8/94.6
	CEG	54.0/82.3	11.7/97.1	57.7/85.7	16.1/96.6	28.7/92.2	4.3/98.2	5.0/97.8
	CEG-P	64.0/79.8	22.4/94.8	83.1/78.4	32.4/92.1	34.4/91.1	74.9/75.5	17.6/95.5
MaxLogit	Raw	75.7/76.9	14.5/96.2	61.6/85.7	23.4/95.1	44.8/89.5	3.0/98.7	5.2/98.1
	CEG	54.5/82.3	11.2/97.1	52.7/87.5	15.3/97.1	30.4/92.0	3.2/98.5	4.7/98.0
	CEG-P	71.5/78.7	13.2/96.6	59.1/86.5	22.0/95.7	40.2/90.4	3.0/98.6	5.0/98.1
Energy	Raw	97.1/41.2	68.3/70.5	76.7/64.4	34.0/90.3	94.9/50.0	19.8/93.7	52.0/86.0
	CEG	58.0/79.4	12.8/96.8	55.5/86.9	15.5/97.0	34.2/90.6	4.1/98.2	4.9/97.9
	CEG-P	96.7/50.7	65.8/77.7	75.7/71.1	31.9/92.3	93.3/60.5	16.8/94.9	44.1/89.4
MCM	Raw	60.7/81.9	19.3/94.7	82.8/76.3	31.5/93.4	34.8/90.9	26.1/94.1	8.6/97.2
	CEG	53.9/82.9	11.0/97.2	57.4/85.8	15.3/97.0	29.5/92.2	4.0/98.4	4.9/97.9
	CEG-P	58.7/82.6	16.6/95.5	79.5/79.2	27.2/94.5	32.9/91.5	21.4/95.2	7.6/97.4
GL-MCM [†]	Raw	53.4/83.2	16.4/95.3	68.1/79.8	31.4/92.9	30.2/91.6	49.1/85.6	5.6/98.0
	CEG	52.8/82.9	10.8/97.1	53.7/86.4	13.9/97.3	28.5/92.2	4.1/98.3	4.8/98.0
	CEG-P	52.6/83.4	14.7/95.9	65.7/81.9	28.9/94.2	29.2/91.9	43.8/88.8	5.4/98.0
Remaining domains and equal-unit mean								
Base	Variant	Pets	Caltech	EuroSAT	DTD	VOC	DomainNet	Mean
MSP	Raw	37.8/91.3	32.6/91.2	99.2/31.0	83.3/71.4	47.5/84.2	90.0/61.8	57.2/78.5
	CEG	0.5/99.7	27.2/93.3	65.1/67.3	58.2/87.6	41.1/88.9	84.9/67.1	35.0/88.8
	CEG-P	33.2/93.6	30.9/92.0	99.0/41.0	81.1/76.2	47.5/85.7	89.8/63.6	54.6/81.5
MaxLogit	Raw	0.4/99.9	45.5/88.3	57.2/69.3	64.5/85.4	53.8/84.3	87.6/56.7	41.3/86.5
	CEG	0.4/99.7	28.8/92.7	57.8/70.6	58.1/88.1	38.5/89.3	84.0/66.1	33.8/89.2
	CEG-P	0.4/99.8	42.5/89.7	56.9/69.8	61.5/86.7	51.9/85.8	87.4/59.7	39.6/87.4
Energy	Raw	18.1/95.0	95.1/49.1	16.8/95.9	74.9/78.6	91.0/54.3	84.6/38.9	63.3/69.8
	CEG	0.4/99.7	33.6/91.2	24.1/93.5	58.2/87.4	43.6/87.7	79.6/64.0	32.7/90.0
	CEG-P	14.8/96.3	94.0/60.3	17.5/95.6	71.7/82.0	89.4/62.9	84.5/44.7	61.2/75.2
MCM	Raw	7.5/98.4	37.5/91.1	99.6/20.2	83.4/74.2	44.6/86.0	90.3/63.8	48.2/81.7
	CEG	0.5/99.7	28.0/93.1	65.0/67.3	57.7/88.1	41.3/89.1	85.1/68.6	34.9/89.0
	CEG-P	5.7/98.7	33.7/91.8	99.5/33.3	80.8/78.4	44.2/87.0	90.3/65.5	46.0/83.9
GL-MCM [†]	Raw	10.8/97.7	34.5/92.2	93.1/50.3	64.8/83.8	37.2/89.1	87.8/65.6	44.8/85.0
	CEG	0.5/99.7	28.2/93.2	64.2/69.0	53.1/88.9	36.2/89.8	84.6/68.4	33.5/89.3
	CEG-P	8.5/98.2	31.0/92.6	91.2/55.6	62.7/85.5	35.2/89.4	88.0/66.7	42.8/86.3

Table F3: **Full all-domain CEG audit: SigLIP2-B/16**. Each cell is FPR95↓/AUROC↑ on every image; Raw is the base score, CEG is the non-compensatory minimum, and CEG-P is the protected blend. For guarded rows, green/red denotes FPR95 lower/higher than Raw by more than 0.10 points and gray denotes a practical tie; color is keyed to FPR95 only. Values average OOD sources within a task; DomainNet averages its six styles once. This is post-result development evidence.

Native and natural/fine-grained units								
Base	Variant	IN-1K	C-10	C-100	MNIST	IN-200	CUB	Food
MSP	Raw	56.5/83.6	36.1/93.5	85.9/77.4	88.1/83.1	23.4/94.3	78.3/77.6	32.5/93.6
	CEG	58.6/84.0	18.1/95.4	59.3/84.2	53.4/88.0	23.9/93.6	2.9/98.8	6.7/97.5
	CEG-P	55.9/84.4	31.3/94.3	83.5/79.7	85.8/84.8	22.4/94.4	72.3/82.8	27.2/94.7
MaxLogit	Raw	55.8/84.9	8.0/97.6	39.7/90.9	10.0/97.9	25.6/93.5	2.0/99.2	3.8/98.4
	CEG	58.7/84.3	10.4/96.9	44.9/88.8	12.5/97.5	25.4/93.3	2.1/99.1	4.2/98.2
	CEG-P	55.3/85.2	8.1/97.6	38.9/90.8	9.5/98.0	24.4/93.7	1.9/99.2	3.8/98.4
Energy	Raw	95.8/47.1	49.9/77.6	73.8/65.5	11.8/97.4	94.6/52.1	30.2/91.8	15.4/95.2
	CEG	69.5/79.4	20.8/94.7	56.3/85.2	13.7/97.3	38.3/89.6	2.8/98.8	6.4/97.7
	CEG-P	95.2/54.7	47.1/82.0	72.8/70.9	11.4/97.6	92.9/61.6	26.5/93.7	12.4/96.0
MCM	Raw	52.1/86.7	16.6/95.2	85.7/78.8	88.9/82.4	19.2/94.8	21.2/95.7	11.1/96.5
	CEG	58.5/84.5	16.8/95.7	58.5/84.3	53.3/87.8	22.6/93.9	2.6/98.9	6.3/97.6
	CEG-P	51.7/86.6	14.9/95.6	83.5/80.7	87.2/84.1	19.1/94.8	16.7/96.5	9.3/96.9
GL-MCM [†]	Raw	55.8/85.5	25.7/94.3	83.0/76.6	83.3/82.0	25.7/93.5	27.5/94.3	13.5/96.2
	CEG	59.3/84.4	18.2/95.4	57.1/84.5	52.9/87.5	27.3/93.0	2.6/98.9	6.4/97.6
	CEG-P	55.3/85.8	20.9/94.8	80.3/79.2	81.3/83.7	25.0/93.7	22.4/95.5	10.5/96.6
Remaining domains and equal-unit mean								
Base	Variant	Pets	Caltech	EuroSAT	DTD	VOC	DomainNet	Mean
MSP	Raw	46.2/91.9	19.4/94.9	99.5/24.5	70.3/81.1	57.2/84.4	83.5/67.2	59.8/80.5
	CEG	0.4/99.7	16.8/94.5	75.1/61.3	65.5/85.1	45.3/86.3	86.3/64.9	39.4/87.2
	CEG-P	39.9/94.0	17.0/95.0	99.3/34.5	68.7/82.9	56.1/85.3	83.7/67.2	57.2/82.6
MaxLogit	Raw	0.1/100.0	10.9/96.1	49.1/73.0	71.5/83.7	40.1/86.7	87.8/55.9	31.1/89.1
	CEG	0.1/99.8	11.1/95.5	54.6/71.2	68.0/85.2	43.6/87.1	85.0/64.1	32.4/89.3
	CEG-P	0.1/99.9	10.7/96.0	49.7/72.5	69.9/84.9	40.3/87.1	87.8/58.3	30.8/89.3
Energy	Raw	4.1/98.7	89.7/68.8	9.3/97.9	90.9/63.5	83.2/66.8	84.1/33.3	56.4/73.5
	CEG	0.3/99.8	22.5/93.4	17.4/96.3	73.9/81.1	52.2/84.2	78.9/62.0	34.8/89.2
	CEG-P	3.3/98.9	86.7/75.5	10.0/97.7	89.8/69.1	80.9/71.6	84.1/38.4	54.9/77.5
MCM	Raw	7.3/98.3	12.5/95.6	99.7/13.4	64.6/86.1	50.7/86.7	86.1/65.9	47.4/82.8
	CEG	0.3/99.8	13.5/95.1	75.1/61.0	64.4/86.2	45.4/86.9	87.4/64.9	38.8/87.4
	CEG-P	5.3/98.7	12.0/95.6	99.6/27.2	63.2/86.8	49.0/87.0	86.3/66.1	46.0/84.4
GL-MCM [†]	Raw	14.2/97.0	18.7/94.3	99.9/11.0	54.0/89.0	43.6/87.2	86.2/65.6	48.5/82.0
	CEG	0.3/99.8	17.3/94.3	74.9/61.8	61.3/87.6	44.9/86.8	87.5/64.2	39.3/87.4
	CEG-P	9.9/97.8	16.8/94.5	99.9/26.1	53.2/89.3	45.1/87.3	86.3/65.7	46.7/83.8

Table F4: **Full all-domain CEG audit: PE-Core-B/16**. Each cell is FPR95↓/AUROC↑ on every image; Raw is the base score, CEG is the non-compensatory minimum, and CEG-P is the protected blend. For guarded rows, green/red denotes FPR95 lower/higher than Raw by more than 0.10 points and gray denotes a practical tie; color is keyed to FPR95 only. Values average OOD sources within a task; DomainNet averages its six styles once. This is post-result development evidence.

Native and natural/fine-grained units								
Base	Variant	IN-1K	C-10	C-100	MNIST	IN-200	CUB	Food
MSP	Raw	64.6/79.4	13.3/96.7	49.0/87.2	48.5/85.5	35.4/92.1	38.7/90.9	20.3/95.0
	CEG	53.6/84.4	11.0/97.1	43.3/88.8	10.6/97.4	32.1/92.0	3.3/98.8	5.7/97.8
	CEG-P	62.5/81.3	11.7/96.9	46.9/88.1	44.0/88.6	33.3/92.5	33.6/93.0	17.5/95.8
MaxLogit	Raw	60.0/84.2	12.2/96.8	32.5/93.0	3.0/99.2	28.6/93.3	1.8/99.3	3.6/98.4
	CEG	53.3/85.3	11.5/96.9	34.4/92.1	3.8/99.0	29.5/92.7	2.0/99.1	3.8/98.3
	CEG-P	58.3/85.0	11.0/97.0	31.2/93.1	2.9/99.2	27.6/93.4	1.8/99.2	3.6/98.4
Energy	Raw	97.2/37.0	69.0/54.3	75.9/53.8	3.2/99.3	95.9/38.9	44.0/90.2	15.5/95.6
	CEG	59.4/80.2	18.2/95.3	50.1/86.7	3.4/99.1	44.8/88.4	3.5/98.8	5.0/98.0
	CEG-P	97.0/46.7	66.4/63.0	74.5/60.7	2.9/99.3	95.2/49.8	36.3/92.6	12.7/96.3
MCM	Raw	51.1/85.7	8.8/97.4	35.3/91.5	37.5/89.0	25.0/93.6	4.9/98.4	7.0/97.3
	CEG	50.7/85.7	9.2/97.3	37.0/90.6	9.5/97.7	27.6/92.8	2.5/98.9	5.2/97.9
	CEG-P	50.2/86.1	8.6/97.4	34.7/91.6	33.3/91.3	24.6/93.6	4.1/98.5	6.6/97.5
GL-MCM [†]	Raw	48.4/86.4	9.7/97.2	39.9/90.3	46.9/86.4	24.2/93.6	5.4/98.2	7.9/97.0
	CEG	50.4/85.6	10.1/97.1	40.1/89.7	11.6/97.2	28.7/92.6	2.7/98.9	5.3/97.9
	CEG-P	48.1/86.5	9.5/97.2	39.3/90.4	42.7/89.3	24.2/93.6	4.6/98.4	7.3/97.3
Remaining domains and equal-unit mean								
Base	Variant	Pets	Caltech	EuroSAT	DTD	VOC	DomainNet	Mean
MSP	Raw	25.2/95.9	32.5/91.9	98.7/40.4	63.4/84.4	47.6/86.2	87.9/62.3	48.1/83.7
	CEG	0.4/99.7	26.1/92.4	37.6/87.0	61.2/86.8	39.9/88.5	84.4/66.4	31.5/90.5
	CEG-P	19.4/96.9	30.3/92.4	98.1/52.6	63.1/85.6	45.1/87.1	88.2/63.7	45.7/85.7
MaxLogit	Raw	0.3/99.9	12.4/95.8	37.2/89.2	35.0/93.0	42.6/87.2	83.6/66.7	27.1/92.0
	CEG	0.2/99.8	18.4/94.5	35.8/87.9	49.2/90.6	41.8/88.6	82.7/66.7	28.2/91.6
	CEG-P	0.3/99.9	13.1/95.6	35.7/89.2	35.1/93.0	42.5/87.9	83.4/67.0	26.7/92.1
Energy	Raw	39.8/87.5	67.3/74.2	4.2/99.0	74.8/76.5	96.0/51.6	81.6/47.1	58.8/69.6
	CEG	0.4/99.7	30.2/92.6	17.9/97.5	65.0/85.3	46.9/84.4	78.0/63.5	32.5/90.0
	CEG-P	33.6/90.7	63.4/79.2	4.5/99.0	73.4/80.1	95.3/59.9	81.6/50.5	56.7/74.4
MCM	Raw	1.0/99.8	26.7/93.1	98.9/27.8	45.6/89.9	37.9/88.4	89.4/61.7	36.1/85.7
	CEG	0.3/99.8	26.2/93.0	37.4/87.1	54.1/88.9	38.7/89.2	85.1/66.9	29.5/91.2
	CEG-P	0.8/99.7	24.7/93.3	98.5/43.7	45.4/90.1	36.5/88.9	89.7/63.4	35.2/87.3
GL-MCM [†]	Raw	1.0/99.7	26.5/93.1	99.7/21.8	55.4/87.8	34.7/89.2	89.1/61.5	37.6/84.8
	CEG	0.3/99.8	27.4/92.8	36.8/87.7	57.4/87.7	39.4/89.1	85.0/66.6	30.4/91.0
	CEG-P	0.8/99.7	24.9/93.3	99.6/40.3	54.7/88.2	34.6/89.5	89.4/63.1	36.9/86.7

Table F5: **Unaggregated ID×OOD-source MCM results (part 1/5)**. Each VLM cell is Raw/CEG/CEG-P FPR95↓ (%). Guarded numbers are green/red when lower/higher than Raw by more than 0.10 points and gray for a practical tie. No OOD source or DomainNet style is averaged. Guarded variants average five ID-only calibration seeds.

ID dataset	OOD source	Type	CLIP-B/16	SigLIP2-B/16	PE-Core-B/16
IN-1K	SSB-hard	Near	88.7/85.5/88.1	79.2/82.3/79.2	85.3/84.0/85.1
	NINCO	Near	79.9/72.0/78.7	65.8/70.3/65.4	71.0/69.0/70.1
	iNaturalist	Far	31.9/16.0/27.3	30.5/42.0/29.8	28.0/30.1/26.9
	Textures	Far	59.5/59.9/58.5	54.1/57.9/53.8	42.7/39.6/41.1
	OpenImage-O	Far	43.7/36.1/41.0	30.8/40.1/30.4	28.5/30.8/27.7
CIFAR-10	CIFAR-100	Near	35.6/29.7/33.3	31.8/32.8/30.0	18.2/23.4/18.6
	TinyImageNet	Near	34.4/16.3/29.5	28.4/30.7/27.6	20.1/22.1/20.2
	MNIST	Far	0.8/1.0/0.7	5.3/11.9/3.5	0.1/0.1/0.1
	SVHN	Far	1.6/2.2/1.7	1.2/4.9/1.0	0.1/0.2/0.1
	Places365	Far	29.6/11.1/24.0	25.5/17.8/22.4	11.6/8.1/10.8
CIFAR-100	Textures	Far	13.8/5.6/10.7	7.4/2.8/4.9	2.7/1.4/2.2
	CIFAR-10	Near	88.7/82.8/88.0	87.3/63.7/84.5	52.2/56.8/51.5
	TinyImageNet	Near	93.3/64.8/91.7	88.1/73.4/86.2	63.4/68.1/63.4
	MNIST	Far	61.8/38.6/54.2	100.0/54.7/99.9	8.2/11.4/8.0
	SVHN	Far	63.4/41.4/57.6	46.4/49.6/43.7	0.8/2.0/0.8
	Places365	Far	97.8/57.3/96.3	98.6/71.2/97.6	55.3/49.0/53.3
	Textures	Far	92.0/59.6/89.4	94.1/38.6/89.3	31.7/35.0/30.9

Table F6: **Unaggregated ID×OOD-source MCM results (part 2/5)**. Each VLM cell is Raw/CEG/CEG-P FPR95↓ (%). Guarded numbers are green/red when lower/higher than Raw by more than 0.10 points and gray for a practical tie. No OOD source or DomainNet style is averaged. Guarded variants average five ID-only calibration seeds.

ID dataset	OOD source	Type	CLIP-B/16	SigLIP2-B/16	PE-Core-B/16
MNIST	notMNIST	Near	10.3/8.8/9.4	80.1/64.2/78.5	60.8/32.8/57.6
	FashionMNIST	Near	41.9/31.4/39.1	75.3/71.7/73.8	37.0/3.3/32.7
	Textures	Far	32.4/17.4/26.9	93.7/24.3/90.7	31.0/6.6/26.0
	CIFAR-10	Far	28.0/19.6/25.0	96.6/49.4/95.5	23.9/5.4/20.7
	TinyImageNet	Far	30.1/4.6/23.4	90.5/52.2/88.5	28.4/5.5/24.7
	Places365	Far	46.4/10.0/39.2	97.2/58.0/96.4	43.7/3.1/38.4
IN-200	SSB-hard	Near	61.1/53.6/58.7	37.1/41.5/36.9	48.2/49.6/47.7
	NINCO	Near	53.3/46.3/51.4	29.8/35.2/29.8	38.5/41.6/38.0
	iNaturalist	Far	6.0/3.5/4.8	3.2/5.5/3.3	4.9/8.7/4.8
	OpenImage-O	Far	18.7/14.7/16.7	6.7/8.4/6.3	8.3/10.3/8.0
CUB-200	SSB-hard	Near	47.2/14.6/41.7	36.3/9.5/31.0	16.5/9.3/14.6
	NINCO	Near	29.9/0.3/23.7	19.9/0.3/14.8	1.8/0.3/1.1
	iNaturalist	Far	10.5/0.5/7.5	6.3/0.2/4.1	0.1/0.3/0.1
	OpenImage-O	Far	16.9/0.5/12.9	22.5/0.3/16.9	1.0/0.2/0.7
Food-101	SSB-hard	Near	12.9/6.1/11.7	14.3/6.3/12.1	11.4/6.2/10.7
	NINCO	Near	16.6/11.1/14.7	22.8/14.7/19.4	13.1/12.0/12.5
	iNaturalist	Far	0.9/0.2/0.7	2.8/1.6/2.0	0.8/0.7/0.8
	OpenImage-O	Far	4.0/2.2/3.4	4.5/2.5/3.6	2.8/1.8/2.4

Table F7: **Unaggregated ID×OOD-source MCM results (part 3/5)**. Each VLM cell is Raw/CEG/CEG-P FPR95↓ (%). Guarded numbers are green/red when lower/higher than Raw by more than 0.10 points and gray for a practical tie. No OOD source or DomainNet style is averaged. Guarded variants average five ID-only calibration seeds.

ID dataset	OOD source	Type	CLIP-B/16	SigLIP2-B/16	PE-Core-B/16
Pets	SSB-hard	Near	15.0/ 1.2/12.3	8.7/1.0/6.8	2.4/0.7/1.8
	NINCO	Near	8.9/0.6/6.9	3.8/0.2/2.3	1.6/0.3/1.3
	iNaturalist	Far	2.0/0.0/1.1	7.8/0.0/5.7	0.0/0.0/0.0
	OpenImage-O	Far	4.0/0.1/2.8	8.9/0.1/6.6	0.1/0.0/0.1
Caltech-101	SSB-hard	Near	60.0/ 42.9/54.4	16.8/ 17.4/16.1	36.9/29.3/33.2
	NINCO	Near	54.9/ 43.5/50.8	26.2/ 27.4/25.4	44.6/ 45.6/42.9
	iNaturalist	Far	10.3/7.8/8.3	1.7/3.2/1.6	11.3/ 14.4/10.0
	OpenImage-O	Far	24.8/ 17.8/21.1	5.2/6.1/4.9	13.9/ 15.4/12.8
EuroSAT	SSB-hard	Near	99.8/54.7/99.7	99.8/59.2/99.8	98.9/10.9/98.4
	NINCO	Near	99.8/44.2/99.7	99.5/67.1/99.3	98.7/22.7/98.1
	iNaturalist	Far	100.0/99.0/100.0	100.0/99.8/100.0	100.0/79.0/100.0
	OpenImage-O	Far	98.9/62.3/98.6	99.4/74.4/99.3	98.0/37.0/97.4
DTD	SSB-hard	Near	89.7/54.0/87.0	64.4/62.7/62.8	50.2/55.0/49.8
	NINCO	Near	86.7/60.4/84.6	67.6/65.8/66.4	54.2/64.7/54.7
	iNaturalist	Far	76.0/55.0/72.8	66.2/69.6/65.0	31.1/45.2/31.1
	OpenImage-O	Far	81.3/61.6/78.7	60.3/59.6/58.7	46.9/51.5/45.9

Table F8: **Unaggregated ID×OOD-source MCM results (part 4/5)**. Each VLM cell is Raw/CEG/CEG-P FPR95↓ (%). Guarded numbers are green/red when lower/higher than Raw by more than 0.10 points and gray for a practical tie. No OOD source or DomainNet style is averaged. Guarded variants average five ID-only calibration seeds.

ID dataset	OOD source	Type	CLIP-B/16	SigLIP2-B/16	PE-Core-B/16
VOC-13	SSB-hard	Near	76.8/73.1/76.4	72.2/67.7/71.2	67.8/67.1/67.3
	NINCO	Near	48.5/43.9/48.3	49.3/41.1/47.9	35.6/40.0/35.2
	iNaturalist	Far	12.7/10.9/11.9	43.3/41.9/41.3	20.2/20.5/17.4
	OpenImage-O	Far	40.4/37.3/40.1	37.8/30.9/35.4	28.0/27.1/26.0
DN-real	SSB-hard	Near	83.2/87.1/83.6	69.6/76.7/70.2	81.5/83.1/81.5
	NINCO	Near	72.2/76.5/72.3	54.9/67.0/56.1	68.6/77.6/69.5
	iNaturalist	Far	60.0/77.1/62.0	79.6/92.1/81.8	71.6/87.8/74.6
	OpenImage-O	Far	65.6/73.0/66.2	54.1/62.9/54.7	60.8/68.6/61.3
DN-clipart	SSB-hard	Near	95.6/92.9/95.2	82.8/89.8/83.2	88.9/88.8/89.0
	NINCO	Near	90.8/88.0/90.4	74.5/85.1/75.0	80.9/87.0/82.0
	iNaturalist	Far	93.6/94.2/93.8	97.2/98.9/97.6	91.0/95.4/92.2
	OpenImage-O	Far	88.4/86.2/88.0	71.7/80.6/72.1	74.4/78.8/74.8
DN-painting	SSB-hard	Near	96.9/93.3/96.3	89.5/92.3/89.7	94.1/94.0/94.1
	NINCO	Near	92.7/88.0/92.2	85.8/88.3/85.8	90.6/92.4/90.8
	iNaturalist	Far	96.0/94.2/95.9	99.2/99.4/99.2	98.2/98.2/98.3
	OpenImage-O	Far	90.9/86.6/90.2	80.6/84.2/80.8	84.1/86.3/84.3

Table F9: **Unaggregated ID×OOD-source MCM results (part 5/5)**. Each VLM cell is Raw/CEG/CEG-P FPR95↓ (%). Guarded numbers are green/red when lower/higher than Raw by more than 0.10 points and gray for a practical tie. No OOD source or DomainNet style is averaged. Guarded variants average five ID-only calibration seeds.

ID dataset	OOD source	Type	CLIP-B/16	SigLIP2-B/16	PE-Core-B/16
DN-sketch	SSB-hard	Near	96.5/94.4/96.1	90.8/94.0/91.3	95.2/94.5/95.3
	NINCO	Near	92.1/89.6/91.7	87.9/89.9/87.9	92.5/93.0/92.9
	iNaturalist	Far	95.2/95.4/95.4	99.4/99.5/99.5	98.9/98.4/99.0
	OpenImage-O	Far	90.1/87.8/89.8	82.6/86.2/82.7	86.4/87.7/86.7
DN-quickdraw	SSB-hard	Near	99.7/70.0/99.6	96.5/82.5/96.0	99.2/62.2/99.0
	NINCO	Near	97.6/63.0/97.3	95.2/81.1/94.6	98.4/69.9/98.1
	iNaturalist	Far	99.7/62.4/99.6	99.9/95.0/99.9	100.0/64.1/100.0
	OpenImage-O	Far	97.9/62.3/97.4	90.9/76.0/90.1	96.4/55.6/95.8
DN-infograph	SSB-hard	Near	96.2/97.4/96.2	96.8/96.2/96.7	99.4/96.0/99.3
	NINCO	Near	91.6/93.8/91.9	95.5/92.0/95.1	98.6/94.7/98.5
	iNaturalist	Far	94.7/97.7/95.2	99.9/99.5/99.9	100.0/98.5/100.0
	OpenImage-O	Far	89.5/92.1/89.8	91.3/89.5/91.0	96.9/91.2/96.6